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THE ECONOMIC IMPACT OF DROUGHT ON ALFALFA PRICE FORMATION AND
IRRIGATION MODERNIZATION FEASIBILITY IN THE WESTERN U.S.

by

Saba Ardakani

A thesis submitted in partial fulfillment
of the requirements for the degree

of

MASTER OF SCIENCE

in

Applied Economics

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2025

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ABSTRACT

The Economic Impact of Drought on Alfalfa Price Formation and Wheel line Irrigation
Modernization Feasibility in The Western U.S.

by

Saba Ardakani, Master of Science

Utah State University, 2025

Major Professor: Dr. Ryan Larsen

Department: Applied Economics

This thesis investigates the economic impacts of drought on alfalfa prices and evaluates the feasibility of modernizing irrigation systems for alfalfa production in the Western United States. Utilizing a comprehensive monthly dataset from nine Western states spanning 2021 to 2024, the study employs fixed-effects regression models to quantify the influence of drought severity, forage quality and packaging, export activity, feed market prices, and inventory on alfalfa price formation. The analysis finds that drought severity has a positive correlation with price, especially when compounded by product characteristics. Corn prices also contribute significantly to alfalfa price volatility, while inventory levels play a comparatively limited moderation role. The study further evaluates the economic viability of upgrading wheel-line irrigation systems, using cost-benefit analysis and financial metrics. Modernization offers major savings in water, labor, and energy, while improving yield stability. The results emphasize drought's impact on forage markets and the value of modernizing wheel line irrigation to support sustainable alfalfa production in the Western U.S.

(101 pages)

PUBLIC ABSTRACT

The Economic Impact of Drought on Alfalfa Price Formation and Wheel line Irrigation

Modernization Feasibility in The Western U.S.

Saba Ardakani

This research examines how drought severity affects alfalfa hay prices across nine Western states from 2021 to 2024, using detailed economic analysis and market data. The findings show that droughts significantly raise alfalfa prices, especially when combined with other market and product factors such as forage quality and package, while other factors like corn prices also influence the market. Additionally, the research examines the economic potential of upgrading wheel-line irrigation systems with technologies such as low-pressure sprinklers and automated controls. Modernization substantially reduces water, labor, and energy costs while boosting crop yields, offering a practical solution for farmers. In Utah, widespread adoption could generate significant economic benefits over time. These findings provide clear guidance for farmers and policymakers to strengthen alfalfa production against drought and climate challenges, promoting sustainable agriculture in the Western United States.

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CONTENTS

	Page
ABSTRACT	iii
PUBLIC ABSTRACT	iv
ACKNOWLEDGMENTS.....	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
CHAPTER1: INTRODUCTION.....	1
1.1 Background and Motivation	1
1.2 Research Objectives and Thesis Structure	8
CHAPTER 2: QUANTIFYING DROUGHT IMPACT ON ALFALFA PRICE FORMATION IN WESTERN STATES	11
2.1 Introduction.....	11
2.2 Literature Review: Drivers of Alfalfa Hay Price Dynamics	12
2.2.1 Drought and Environmental Drivers of Alfalfa Prices	12
2.2.2 Forage Quality and Packaging Effects.....	14
2.2.3 Market Signals: Feed, Dairy, and Export Demand	15
2.3 Data and Methods	19
2.3.1 Study Area and Scope	19
2.3.2 Data Source and Variable Construction	19
2.3.3 Empirical Model Specification	21
2.4 Results.....	25
2.4.1 Descriptive Statistics and Trends	26
2.4.2 Role of Product Characteristics	30
2.4.3 Influence of Export Activity and Feed Market Prices	31
2.4.4 Baseline Regression Model Results.....	33

2.4.5	Interaction Model Results	36
2.5	Discussion	44
CHAPTER 3: MODERNIZING WHEEL-LINE IRRIGATION SYSTEMS: A PATH TO SUSTAINABLE ALFALFA PRODUCTION		46
3.1	Introduction	46
3.2	Limitations and Inefficiencies of Wheel-Line Irrigation Systems	51
3.3	Economic Framework for Evaluating Wheel-Line Modernization	54
3.3.1	Cost-Benefit Analysis (CBA) Approach	55
3.3.2	Key Economic Metrics and Their Application	56
3.3.3	Risk Assessment and Sensitivity Analysis	59
3.4	Economic Components of Irrigation Modernization: Costs and Benefits	60
3.4.1	Costs	60
3.4.2	Benefits	61
3.5	Data Collection and Simulation Model	65
3.5.1	Model Structure and Scenario Definition	65
3.5.2	Guidance for Application and Implementation	66
3.6	Economic Value of Upgrading Wheel-Line Irrigation in Utah	67
3.6.1	Traditional and Modernized Wheel-Line Systems	68
3.6.2	Statewide Impact of Wheel-Line Modernization	70
3.7	Potential Challenges and Considerations	73
3.8	Discussion and Summary	74
CHAPTER 4: CONCLUSION		76
4.1	Summary of Findings	76
4.2	Suggestions for Future Work	78
CHAPTER 5: REFERENCES		80

LIST OF TABLES

	Page
Table 1. Summary of Variables, Definitions, Unit, and Source	20
Table 2. Empirical Models Specification, Features and Interaction Terms Included	22
Table 3. Summary Statistics of Key Variables	26
Table 4. Baseline Model Regression Results for Alfalfa Hay Prices, (2021-2024).....	35
Table 5. Interaction Models for Alfalfa Price Determination in The Western U.S.	41
Table 6. Wheel line Irrigation Initial and Operating Costs	60
Table 7. Summary of Per-Acre Performance Metrics for Wheel-Line Modernization.....	68
Table 8. Statewide Impact on Adoption Scenarios and Potential Savings.....	71
Table 9. Different Scenario Outcomes of Modernized Wheel Line.....	72

LIST OF FIGURES

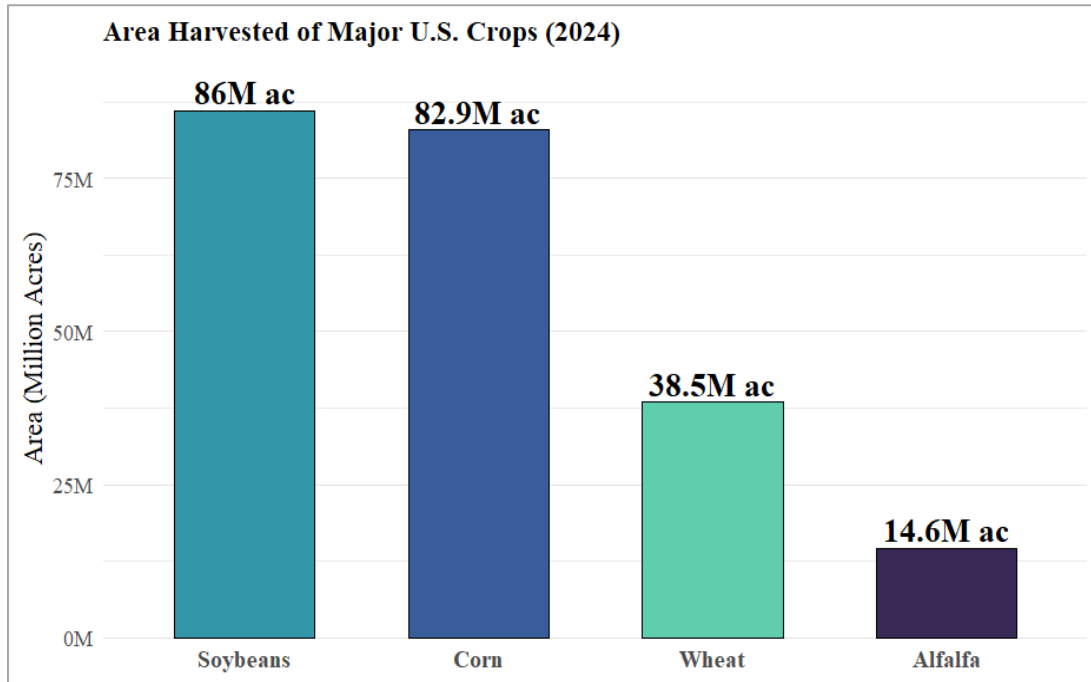
	Page
Figure 1. Area Harvested of Major U.S. Crops.....	2
Figure 2. Production Value of Major U.S. Crops.....	2
Figure 3. Western U.S. States' Share of National Alfalfa Production (1990–2024).....	4
Figure 4. Western U.S. Alfalfa Production Volume Trend Over Time (1990–2024).....	5
Figure 5. Annual Average U.S. Alfalfa Hay Price (1990–2024).....	8
Figure 6. Alfalfa Average Price Over Time (2021 – 2024).....	28
Figure 7. Alfalfa Hay Prices (USD/Ton) Across Western U.S. States (2021 To 2024)	29
Figure 8. Alfalfa Price and Drought Severity Relationship (2021-2024).....	30
Figure 9. Average Alfalfa Hay Packaging and Quality Impact on Price Across Western U.S. States (2021–2024)	31
Figure 10. Alfalfa Price Relationship with Export Volume to China (2021 – 2024).....	32
Figure 11. Monthly Alfalfa Export Quantity to China From 2021 To 2024	33
Figure 12. Total Irrigated Acres by Western State	47
Figure 13. Number Of Farms with Irrigation by Western State.....	48
Figure 14. Total Acres Irrigated Using Wheel Line Systems by State.	49
Figure 15. Number Of Farms Using Wheel Line Irrigation Systems by State	50

CHAPTER1: INTRODUCTION

1.1 Background and Motivation

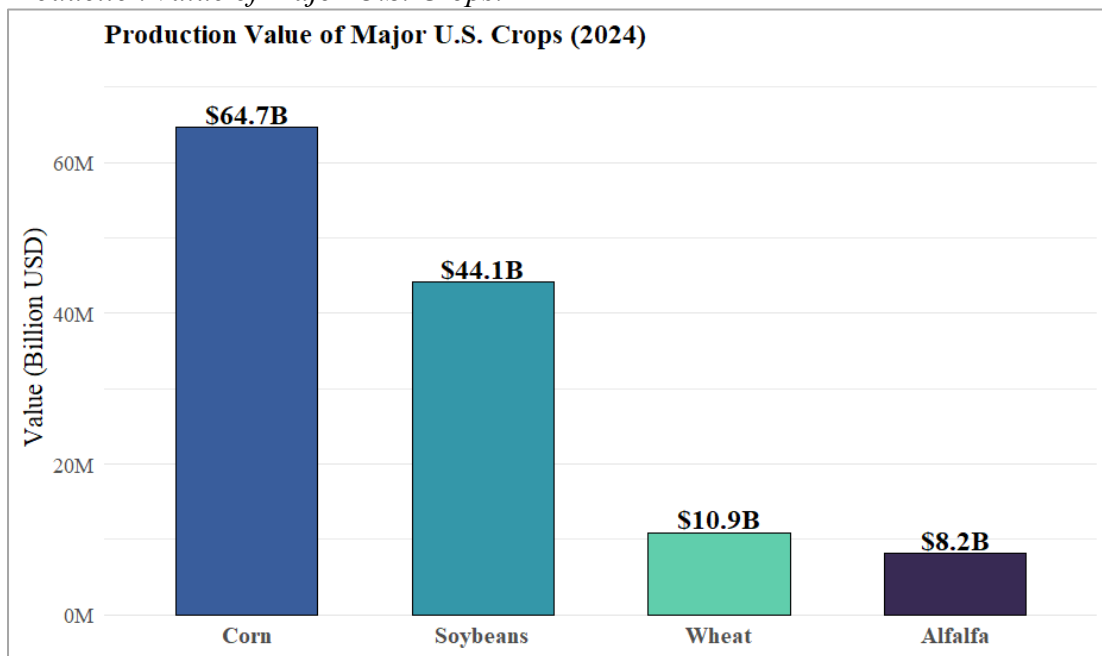
Alfalfa is one of the most widely cultivated forage crops globally and the fourth most valuable field crop in the United States (NAAIC, 2017). It is an energy-efficient crop that provides significant nitrogen fixation, reducing the need for expensive fertilizers and Its cultivation enhances soil structure and improves water infiltration, making it a valuable resource for sustainable agricultural practices (Radovic et al., 2009). As a perennial with deep roots, alfalfa is more resilient to drought than many annual crops, supporting animal husbandry and broader agricultural systems. (NAAIC, 2017). According to the United States Department of Agriculture (USDA) , the production value reached to \$8.16 billion with about 15 million acres in 2024, ranking behind corn, soybeans, and wheat, as illustrated in Figures 1 and 2 (USDA NASS, 2024).

Figure 1.
Area Harvested of Major U.S. Crops



Note. alfalfa harvested area ranks fourth in the U.S.

Figure 2.
Production Value of Major U.S. Crops.

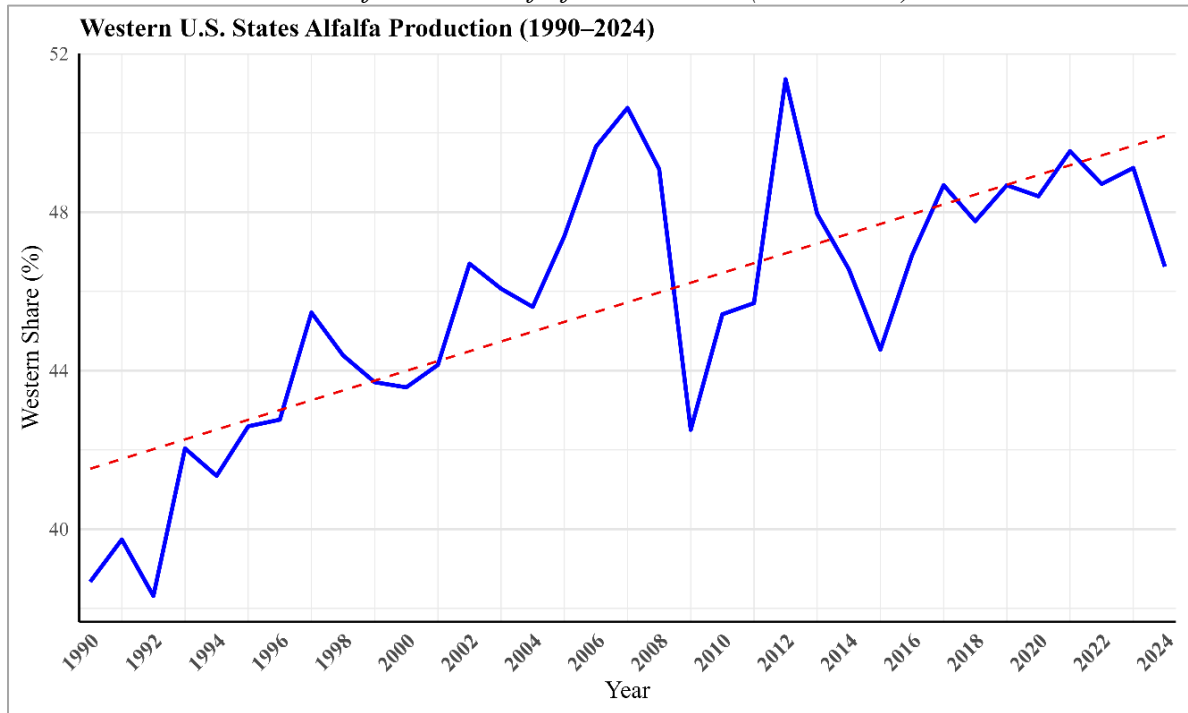


Note. Alfalfa ranks fourth in value in the U.S.

Understanding the regional dynamics of alfalfa production is critical due to its economic and ecological importance. The Western United States has emerged as a dominant center for alfalfa cultivation since the late nineteenth century, with this trend accelerating in recent decades due to favorable agronomic, economic, and climatic conditions (NAAIC, 2017; Putnam et al., 2000). Expansion in states like California, Idaho, and Montana has been driven by increased dairy industry demand, high export activity, and suitable conditions for irrigation (Ottman et al., 2013; Tejeda et al., 2015). In contrast, declines in non-Western states are primarily due to reduced harvested acreage, highlighting the West's growing role in national forage supply (Creech & Foster, 2013).

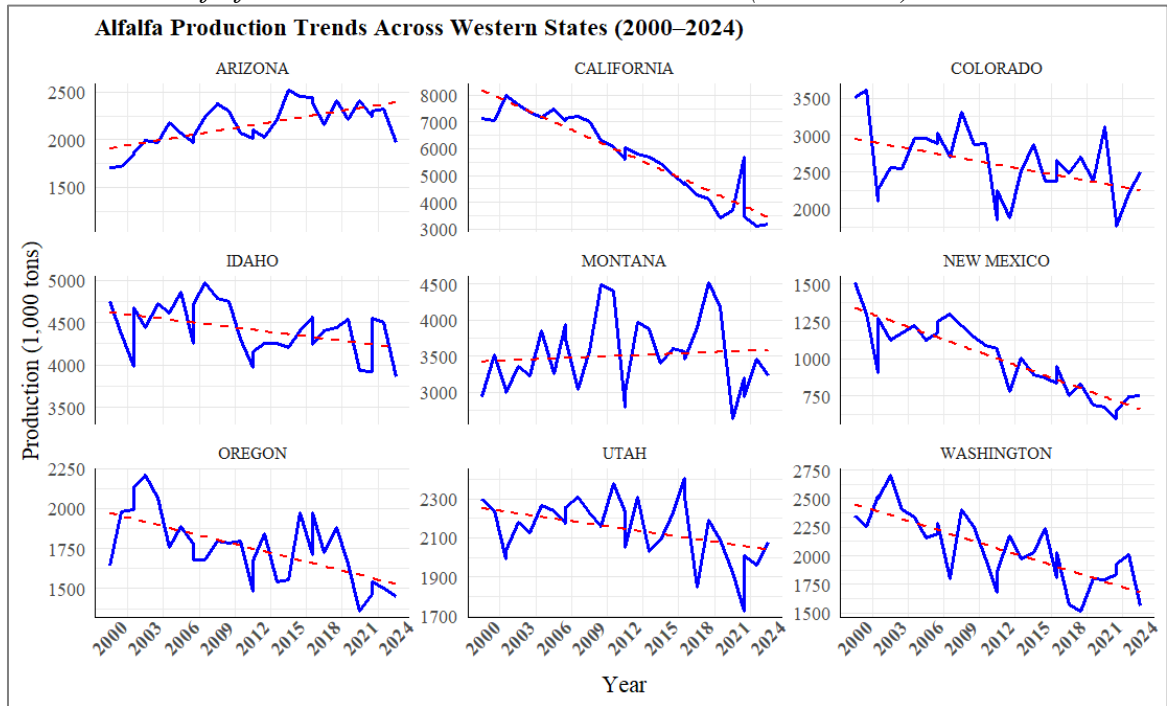
The region's dominance is also evident in export trends. By 2015, nearly 20 percent of all U.S. alfalfa hay was sold internationally, with more than 95 percent of these shipments sourced from Western states (Feuz & Larsen, 2023). The latest data from 2024 indicates that the eleven western states produced over 46 percent of the country's alfalfa, with Idaho, California, and Montana ranking as the leading producers, followed by Colorado, Idaho, and Utah. Meanwhile, other parts of the country have seen declines in production, primarily driven by reductions in harvested acreage (USDA NASS, 2024).

As shown in Figure 3, the annual share of U.S. alfalfa production contributed by western states has exhibited a consistent upward trend, rising from approximately 39% in 1990 to over 50% by the early 2010s. Despite some year-to-year fluctuations, this overall increase underscores the growing significance of the western region within the broader agricultural framework.

Figure 3.*Western U.S. States' Share of National Alfalfa Production (1990–2024)*

Note. The plot shows an overall increasing trend with year-to-year variability.

In addition to the aggregate regional trends, Figure 4 presents production patterns at the state level, revealing greater complexity. While most states display modest year-to-year fluctuations, significant declines are evident in key producing states such as California and New Mexico. In contrast, states like Arizona and Montana have sustained relatively stable production levels throughout the period. These contrasting production trajectories across the western United States underscore the multifaceted factors influencing alfalfa supply in the region. They further emphasize the necessity for a more detailed analysis of the underlying drivers contributing to price volatility.

Figure 4.*Western U.S. Alfalfa Production Volume Trend Over Time (1990–2024)*

Underlying these production and market trends is the persistent challenge of water scarcity in the Western United States. Recent research highlights how groundwater depletion and climate variability further intensify water scarcity risks, underscoring the economic value of conservation and coordinated management to sustain long-term agricultural productivity (Suter et al., 2021).

Over the past two decades, the region has experienced increasingly severe and frequent droughts, driven by a combination of reduced precipitation and rising temperatures linked to anthropogenic climate change (Williams et al., 2020). Severe and frequent droughts in the Western United States have significantly reduced water availability, increasing risks for ecosystems and irrigation-based agriculture. Assessments of major river basins highlight growing drought frequency and severity, emphasizing regional water vulnerability

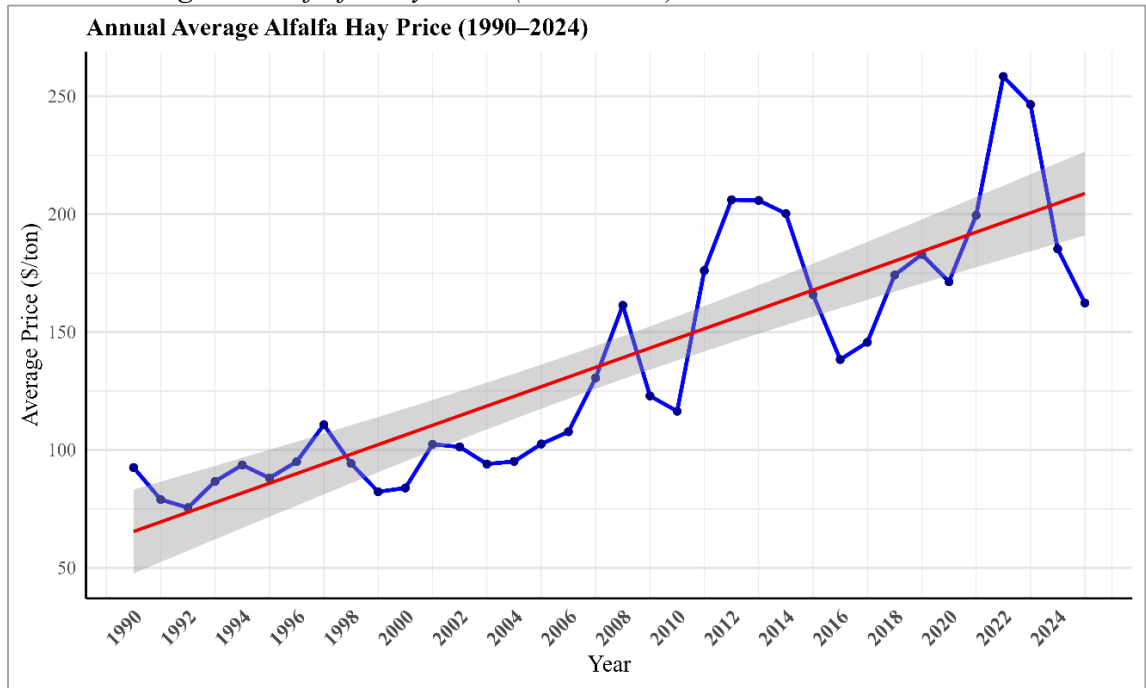
(Shokoohi & Morovati, 2015). Several recent studies have quantified the broader economic repercussions of drought for agriculture in the western United States, emphasizing impacts on crop yields, forage supply, and downstream market volatility (Pritchett et al., 2013).

Recent research shows that, Since 2000, temperature-driven increases in evaporative demand have been the primary factor intensifying drought severity, contributing over 60% to the 2020–2022 drought impact (Williams et al., 2020). Climate projections indicate that without effective mitigation measures, severe drought events like those recently experienced are projected to occur with greater frequency by the mid to late 21st century. This trend underscores the critical need to implement adaptive water management and agricultural strategies across the western United States (Udall & Overpeck, 2017; Williams et al., 2020). Forage crops like alfalfa, while more drought-tolerant than many annual crops, remain highly dependent on irrigation for optimal productivity and quality (Maghsoodi et al., 2017). As a result, the sustainability of alfalfa production in the West is closely linked to the availability and efficient management of water resources.

Beyond production volumes, Western alfalfa markets are shaped by considerable price volatility influenced by a combination of local and global forces (Tejeda et al., 2015). USDA analyses and market reports reveal that alfalfa hay prices have become increasingly sensitive to a diverse array of agricultural practices and broad economic conditions, experiencing significant fluctuations due to fluctuating market dynamics (McCulloch et al., 2014). For instance, the COVID-19 pandemic disrupted supply chains and labor, causing agricultural output losses of 1.18% to 7.14% across U.S. counties, particularly impacting small, non-white, and female-operated farms (Haqiqi & Bahalou Horeh, 2021). Additionally, The 2022 drought reduced hay inventories to their lowest since 2013, increasing cattle feed costs, but

a recovery began in 2023, especially for non-alfalfa hay, with continued improvement projected through 2024 due to better weather (Burdine, 2024). The USDA's 2025 Crop Production report projects further growth in hay production, with a total of 49.39 million acres estimated for harvest in 2025, 14.61 million acres in alfalfa and 34.78 million acres in other hay. However, given the localized nature of hay markets and high transportation costs, the benefits of increased supply may not be distributed evenly across all regions (USDA NASS, 2025b). USDA data analysis, as shown in Figure 5, shows a steady rise in alfalfa hay prices from 1990 to 2024, with notable spikes during economic and environmental crises (USDA NASS, 2025b). Prices experienced substantial increases from 2008 to 2014, and again in the early 2020s following the COVID-19 pandemic and the severe drought of 2022, reflecting the effects of rising input costs, labor disruptions, and constrained hay supply (USDA NASS, 2025b). According to National Oceanic and Atmospheric Administration (NOAA), the 2022 drought was the most widespread in the U.S. since 2012, with more than half the country experiencing moderate to extreme drought conditions. Western states faced particularly severe impacts, with major reservoirs dropping to their lowest levels on record and forage production declining sharply (NOAA, 2023). Despite fluctuations, the regression line in Figure 5 confirms a long-term upward trend in average prices, reflecting persistent market pressures.

Figure 5.
Annual Average U.S. Alfalfa Hay Price (1990–2024)



Note. The plot highlights a long-term upward trend and increased alfalfa price volatility after 2010.

1.2 Research Objectives and Thesis Structure

The primary objective of this thesis is to provide a comprehensive, multi-dimensional analysis of alfalfa hay price formation in the Western United States under drought and water scarcity conditions, while also establishing a foundation for evaluating the feasibility of modernized irrigation systems. It seeks to address critical gaps in the literature by integrating high-frequency, multifactor data to explicitly model the complex interactions between environmental stressors, market dynamics, and supply-side variables. The study aims to quantify the joint influence of drought severity, forage quality, packaging type, export activity (with a particular focus on China-bound shipments), feed market prices, and inventory levels on monthly alfalfa hay prices across nine Western U.S. states. Using a

monthly, multi-state panel data framework, the study quantifies the impacts of drought severity, forage quality, packaging type, export demand (notably to China), feed market prices, and inventory levels on alfalfa prices across nine Western states. It explores how these factors influence price volatility, providing insights into market responses to environmental and economic shocks. The thesis also assesses the costs, benefits, and adoption potential of water-efficient irrigation technologies, such as low-pressure sprinklers and automated systems, to inform producers and policymakers.

This thesis progresses logically from analyzing drought and water scarcity impacts on alfalfa price formation to evaluating economic strategies for adaptation. Chapter 2 reviews the relevant literature and empirical methods used to explore how environmental and market factors influence alfalfa prices across nine Western U.S. states. The chapter presents findings that identify key drivers of price volatility and discusses their implications for producers, policymakers, and other stakeholders. Building on this, Chapter 3 evaluates the economic feasibility and potential benefits of modernizing wheel line irrigation systems, drawing on recent field research and policy analyses to inform future directions. Chapter 4 integrates findings, highlighting sustainable agriculture and future research directions. This research is supported by a Utah State University-led project, funded by the USDA National Institute of Food and Agriculture (NIFA), which optimizes wheel-line irrigation to enhance profitability for small- and medium-sized farms in Utah. The project conducts field trials in Cache, Iron, Duchesne, and Uintah Counties, testing technologies like low-pressure sprinklers, solar-powered movers, and automated shutoff valves to reduce labor, water, and energy costs while improving yields. Currently, detailed data from these trials is not yet available. However, once collected, they will provide a critical empirical foundation for the

economic analyses developed in this thesis. By combining empirical field research with economic modeling, the study aims to provide actionable insights for producers and policymakers regarding the effectiveness and potential benefits of irrigation modernization in drought-prone agricultural regions.

CHAPTER 2: QUANTIFYING DROUGHT IMPACT ON ALFALFA PRICE FORMATION IN WESTERN STATES

2.1 Introduction

Drought has emerged as a defining factor for agricultural resilience in the western United States, with recent decades marked by both increased severity and frequency of water shortages. The interplay of reduced precipitation and sustained warming has not only intensified drought episodes but also fundamentally altered the hydrological landscape, as evidenced by declining reservoir levels and persistent water deficits across major agricultural regions (Williams et al., 2022). Recent economic analyses demonstrate that drought intensity is strongly associated with disruptions in regional forage and feed markets (Koontz, 2025). The southwestern U.S., including California and the Four Corners region, has experienced statistically significant increases in temperature-dominated droughts, which have contributed to persistent water shortages, reduced agricultural productivity, and heightened wildfire risk (Udall & Overpeck, 2017).

This chapter quantifies the impact of drought severity and market factors on alfalfa hay price formation across nine Western U.S. states from August 2021 to September 2024, building on Chapter 1's context of alfalfa's economic significance (46% Western production share) and drought-driven volatility. Using fixed-effects regression models and a monthly, multi-state dataset, the analysis examines how drought severity (Drought Severity and Coverage Index, DSCI), forage quality, packaging type, export volumes to China, feed market prices (corn, feeder cattle), and inventory levels jointly influence alfalfa prices. The findings highlight drought's role in price volatility, informing the irrigation modernization framework in Chapter 3 and policy recommendations in Chapter 4.

2.2 Literature Review: Drivers of Alfalfa Hay Price Dynamics

2.2.1 Drought and Environmental Drivers of Alfalfa Prices

The USDA reports that climate change significantly impacts agricultural productivity, especially in the western U.S., where water scarcity and more frequent droughts pose serious challenges to farming and crop yields (Rowley, 2023; USDA, 2013). These conditions can lead to higher production costs, as farmers must adopt advanced irrigation systems or face reduced yields (Shewmaker et al., 2011). Drought reduces alfalfa dry matter yields and crude protein by, while raising acid detergent fiber (ADF) levels, which limits its suitability for dairy markets requiring a Relative Feed Value (RFV) above 150 (Liu et al., 2017). It restricts supply and heightens competition among livestock producers for forage. Research shows that regional and national droughts can raise hay prices by up to 15%, with effects starting about three months later and lasting nearly a year. Feed grain prices remain stable due to international trade and market integration (Schaub & Finger, 2020). In the short term, drought raises crops, forage prices and lowers cattle prices due to increased herd liquidation. Long-term droughts lead to higher crop prices, reduced livestock inventories, fewer animals in the supply chain, and ultimately higher livestock prices (Leister et al., 2015; Newell, 2023)

Recent studies have shown that drought's impact on hay prices is both immediate and persistent. For example, severe and exceptional drought conditions can increase alfalfa prices by \$16–34 per ton, with effects magnified in regions dependent on rain-fed production (Rowley, 2023). These price shocks are not isolated, and they often cause herd liquidation, reduce livestock inventories, and contribute to longer-term market instability

(Rodziewicz et al., 2023). Moreover, drought interacts with other environmental variables such as precipitation and temperature, with heatwaves compounding water stress and further reducing crop yields (G. Jefferson, 2016; USDA, 2013). The Drought Severity and Coverage Index (DSCI) and related metrics have become essential for quantifying drought impacts at state and district levels, providing a subtle understanding of how water scarcity translates into market shocks (USDA NASS, 2024).

While drought is important, other weather events also play a role. Unlike globally traded crops (e.g., wheat), alfalfa's price sensitivity to local climate reflects its limited storability and high transport costs, which isolate regional markets (Fox et al., 2011; Schaub & Finger, 2020)

Excessive precipitation during critical harvest periods can adversely affect the quality of alfalfa hay, making it unsuitable for premium markets like dairy feed. For example, states like Nevada, Utah, and Idaho face rain-related challenges during harvest that reduce yield and forage quality, while California's more favorable weather supports higher-quality hay and greater market stability. (Hoyt, 2006).

Rainfall affects hay quality during harvest, but temperature patterns consistently influence plant growth and productivity throughout the season (G. Jefferson, 2016). Rising temperatures pose potential challenges to regional water availability, and the water elevation is sensitive to increases in temperatures due to changes in streamflow, precipitation, and evaporation (Affram et al., 2025). Moreover, as global temperatures increase, precipitation becomes a less reliable predictor of hay yields because severe heat induces physiological stress that impairs plant development and lowers productivity (G. Jefferson, 2016). The USDA also notes that higher temperatures can trigger stress responses in alfalfa, reducing

its yield potential, while extreme cold can similarly hinder growth and disrupt optimal harvest timing (USDA, 2013). This is especially critical in the Western United States, where alfalfa production follows strict schedules that are highly influenced by seasonal climate variations (Orloff et al., 2002).

2.2.2 Forage Quality and Packaging Effects

Alfalfa quality is a key determinant of market price, particularly in premium markets such as dairy feed and is typically measured by metrics such as RFV and Total Digestible Nutrients (TDN), which reflect the crop's nutrient density, digestibility, and suitability for different livestock applications (Putnam et al., 2000). Studies have shown that alfalfa with higher protein levels and greater digestibility consistently earns price premiums. Additionally, a higher RFV, which is closely associated with a greater leaf-to-stem ratio and improved digestibility, is a reliable indicator of higher market value (Hopper et al., 2004; Pardew, 1988).

Beyond quality, market preferences related to packaging type also influence price variability. Bale characteristics, particularly size and type, significantly impact pricing. Research on hay auctions reveals that Small Square Bales generally command higher prices, while larger bales are often discounted (McCulloch et al., 2014). An increase in the use of compressed bales and cubes for alfalfa exports to Pacific nations has been observed, highlighting a growing premium market for high-quality, compactly packaged hay, while early harvesting and efficient packaging have been shown to improve market value, especially for export purposes (Eldredge et al., 2001). However, empirical studies quantifying the effect of packaging type on price formation, particularly in the context of

drought, remain limited. By incorporating both forage quality and packaging variables, this study addresses a critical gap in literature. Beyond environmental and product-specific factors, broader feed market signals and supply chain dynamics play a crucial role in shaping alfalfa hay prices.

2.2.3 Market Signals: Feed, Dairy, and Export Demand

Alfalfa hay prices in the Western United States are strongly influenced by a complex set of market signals originating from the feed, dairy, and export sectors. These forces interact with supply shocks such as drought, amplifying or moderating price volatility.

Among the key drivers of alfalfa demand, the dairy industry stands out as the most influential sector, consuming the majority of alfalfa hay in many western states, with up to 75-85% of alfalfa in states like California used in dairy rations (Tejeda et al., 2019). Structural changes in the dairy sector, such as herd expansion or contraction in response to milk price volatility, have direct consequences for regional alfalfa demand and price formation. Simulations of Utah dairy operations, for example, demonstrate that the optimal forage mix-and thus alfalfa demand-shifts in response to changing milk and feed prices (Cann, 2014). Tighter supply conditions can lead to more pronounced price changes. California showed stronger hay price responsiveness to milk prices than Idaho and New Mexico, as evidenced by a higher price responsiveness parameter (Hatzenbuehler et al., 2021). It has been shown that during periods of low milk prices, high-quality alfalfa is often substituted with alternative feed options by dairy producers to control costs, while rising milk prices encourage the use of premium-grade alfalfa, thereby increasing demand and prices(Butler, 2005).

Beyond the direct influence of the dairy industry, the price of substitute feeds, particularly corn, plays a pivotal role in alfalfa price formation. Numerous studies have shown that increases in corn prices lead to higher alfalfa prices, as livestock and dairy producers substitute alfalfa for more expensive grains in their rations (Feuz & Larsen, 2023). Corn, a major component of concentrated animal feeds and a primary energy source, serves as a key substitute for alfalfa hay in many cattle diets (Warner, 2008). High corn prices increase the demand for alfalfa hay as a substitute feed, as livestock producers seek more cost-effective alternatives when grain prices rise. For instance, corn price increases have been shown to significantly influence alfalfa prices, with models explaining up to 71% of alfalfa price variability, suggesting a strong substitution effect in feed markets (Feuz & Larsen, 2023b; Tejeda et al., 2019). Regression analyses across seven Western states indicate that a one standard deviation increase in corn price can raise alfalfa prices by \$15.41 per ton, the largest marginal effect among feed variables (Feuz & Larsen, 2023). Moreover, fluctuations in feed prices have significant implications for livestock producers' profitability. According to NDSU Extension, feed prices for cattle, including alfalfa, have risen sharply in recent years, directly increasing the cost of gain for feedlot operations (NDSU Agriculture, 2023). This dynamic feeds back into demand for alfalfa, especially during periods of high grain prices or poor pasture conditions.

The growing significance of export markets, particularly China, has added a new dimension to alfalfa price dynamics alongside domestic feed market signals. As both a crucial livestock feed and export commodity, alfalfa markets are highly sensitive to shifts in supply and demand (Cann, 2014). China's expanding dairy industry presents substantial opportunities for U.S. alfalfa and dairy exports, although competition from other countries is increasing.

Currently, about 20% of U.S. alfalfa production is exported (Tronstad & Parsons, 2023), with export volumes from the Western United States, especially from California, Idaho, and Washington, rising sharply over the past three decades, notably since around 2009 (Tejeda et al., 2019). However, export markets carry certain risks. While demand from China has been robust due to its limited domestic production and rapid dairy growth, the share of U.S. alfalfa exports to China has declined from over 97% to 63% by early 2019, reflecting China's efforts to diversify forage sources (Q. Wang & Zou, 2020).

According to the USDA Foreign Agricultural Service, in 2023, the U.S. exported 898,186 metric tons of alfalfa hay to China, nearly 90% of China's imports, valued at \$461.7 million, significantly impacting U.S. prices and production. However, exports declined in 2024, with total U.S. hay exports dropping to 3.18 million metric tons and alfalfa shipments to China valued at \$329.76 million (USDA FAS, 2024). Despite the growing impact of China-bound exports on market dynamics, few empirical studies have isolated their unique effects on domestic price formation, especially considering drought conditions and forage quality variation. This study addresses a key gap by explicitly modeling China-specific export volumes alongside drought severity, offering fresh insights into the drivers of regional forage markets.

In addition to export dynamics, other market and supply-side factors including inventory levels further influence alfalfa price formation in the region. Monthly hay inventories, particularly during key seasonal benchmarks, play a significant role in price formation and market dynamics (McCulloch et al., 2014). Alfalfa stocks, especially those measured on May 1, are a critical indicator of supply availability during the peak marketing season. For example, California depends on hay stocks for alfalfa production, which can be 4-5% in

mid-year and 25-30% at the end of the year (Cann, 2014). Lower than normal May 1 inventories are closely associated with increased prices in the second half of the year, while low December 1 inventories constrain early-season supply and drive up prices for the subsequent season (Butler, 2005). Econometric analysis confirms that higher May stocks correlate with lower alfalfa prices across the seven major western states (Feuz & Larsen, 2023). Furthermore, market tightness measured by low stocks-to-use ratios amplifies price responsiveness, particularly in regions like California where inventory volatility is more pronounced (Hatzenbuehler et al., 2021).

These supply-side pressures are further intensified by external market forces such as trade and weather shocks. For instance, an increase in export demand may occur simultaneously with a decline in domestic production due to drought or other unfavorable conditions, leading to sharp price increases. Conversely, periods of overproduction, such as following favorable weather and high acreage, may result in price drops unless offset by strong export or domestic demand (Cann, 2014).

The absence of a centralized futures market for alfalfa hay introduces additional volatility. Unlike crops like corn or soybeans, which have established futures markets, alfalfa prices are determined by localized supply-demand interactions. This market fragmentation makes it challenging to predict prices and manage risk, leading to increased price unpredictability (McCulloch et al., 2014)

Despite substantial research on the environmental and market determinants of alfalfa prices, there remains a lack of integrated studies that simultaneously account for drought severity, forage quality, packaging type, and China-specific, and export activity using high-frequency data. Most prior work has focused on annual or state-level analyses, overlooking the

dynamic, monthly interactions among these factors. This study addresses these gaps by employing a multi-factor, monthly panel framework that incorporates drought severity (DSCI), forage quality, packaging type, export volumes, feed market prices, and inventory levels. By modeling these variables and their interactions, this research provides a more comprehensive understanding of the mechanisms shaping alfalfa price formation in the Western United States.

2.3 Data and Methods

2.3.1 Study Area and Scope

This study focuses on nine Western U.S. states including California, Idaho, Montana, Colorado, Arizona, Utah, New Mexico, Oregon and Washington which together account for the majority of U.S. alfalfa production. These states were selected due to their dominant role in national alfalfa output, their importance in international forage trade, and their vulnerability to climate variability (Putnam et al., 2000). The analysis covers the period from August 2021 to September 2024, capturing recent episodes of extreme drought, recovery, and shifting market dynamics in the region. This range highlights market volatility and climatic stress, notably the 2022 drought and market adjustments. It also reflects post-pandemic disruptions in input costs, labor, and trade, impacting hay pricing and exports.

2.3.2 Data Source and Variable Construction

The empirical analysis uses a comprehensive monthly panel dataset. Alfalfa price, quality, and packaging type were collected from USDA Direct Hay Reports for nine states (USDA - ESMIS 2024). Drought severity is measured using the Drought Severity and Coverage Index (DSCI), a state-level monthly average (index range 0–500) from the U.S. Drought

Monitor. Weather variables include total monthly precipitation (millimeters) and average monthly temperature (°C), both sourced from Climate Engine. Market demand factors are represented by the monthly average near futures price of corn (USD/bushel) and feeder cattle (USD/cwt), with data from USDA NASS and CME Group, reflecting the influence of substitute and complementary feeds in livestock production. Export activity is quantified as total monthly alfalfa hay exports to China (1,000 metric tons), reported by the USDA Foreign Agricultural Service’s Global Agricultural Trade System (GATS), capturing the influence of international demand. Inventory levels are incorporated using state-level alfalfa hay stocks (1,000 tons), reported by USDA-NASS in December and May, to account for seasonal supply fluctuations and market tightness. All variables are harmonized at the state-month level, with state and month fixed effects included to control for unobserved heterogeneity and seasonality. Missing data were addressed through interpolation or exclusion of incomplete observations. Table 1 summarizes the definitions, units, and sources for each variable used in the empirical analysis.

Table 1.
Summary of Variables, Definitions, Unit, and Source

Variable	Definition	Unit/Type	Source
Alfalfa Price	Monthly Alfalfa average price	USD/ton	USDA- ESMIS
DSCI	Drought Severity and Coverage Index	Index (0 –500)	U.S. Drought Monitor
Alfalfa Quality	Alfalfa quality grades	Category	USDA- ESMIS
Packaging Type	Alfalfa Bale types	Category	USDA- ESMIS

Export Volume	Monthly alfalfa export quantity to China	1000 MT	USDA-FAS
Inventory	Alfalfa stocks (December, May)	1,000 Tons	USDA - NASS
Precipitation	Total monthly precipitation	mm	Climate Engine
Temperature	Average monthly temperature	°C	Climate Engine
Corn Price	Nearby futures price of corn	USD/bushel	USDA NASS
Feeder Cattle Price	Nearby futures price of feeder cattle	USD/cwt	USDA NASS
State	State indicator (nine western states)	Categorical	-
Month	Month indicator	Categorical	-

Note. Alfalfa quality grades are reported by USDA which include Premium, Good, Fair, Utility, Supreme. Bale types also reported by UDSA including Large Square 3x4, Large Square 4x4, Small Square, Small Square 3-Tie. <https://www.nass.usda.gov/>

2.3.3 Empirical Model Specification

To evaluate the drivers of alfalfa hay price formation and uncover the mechanisms through which drought, market, and product factors interact, a sequence of fixed-effects linear regression models is estimated. Each model is designed to test a specific hypothesis about how market structure or product differentiation mediates the effect of drought on prices. All models use monthly state-level panel data for nine Western U.S. states, with state-clustered standard errors to account for within-state correlation and heteroskedasticity. To better understand the mechanisms underlying alfalfa hay price dynamics, a series of interaction models is estimated that allow the effect of drought severity on prices to vary according to key market and product characteristics. These interactions provide insight into the pathways through which climate shocks translate into price volatility, highlighting the roles of trade,

market structure, and product attributes. Inventory levels, by contrast, are modeled as a main effect, consistent with their function as a supply buffer rather than a moderator of drought impacts. Table 2 summarizes the structure, variables, and unique focus of each empirical model used in this study.

Table 2.

Empirical Models Specification, Features and Interaction Terms Included

Model Name	Features & Specification	Interaction
Drought-Market Model (Baseline)	Includes all main effects: drought severity (DSCI), precipitation, temperature, ..., with state and month fixed effects	None
Trade Buffer Model	Baseline model plus interaction between drought severity and export volume	Drought $\times$ Export
Market Segmentation Model	Baseline model plus interaction between drought severity and packaging type	Drought $\times$ Packaging
Quality Boost Model	Baseline model plus interaction between drought severity and forage quality	Drought $\times$ Quality
Inventory Model	Baseline model plus inventory levels (May and Dec), without weather variables	Drought $\times$ Stock

(A) Drought-Market Model (Baseline)

The Drought Market Model serves as the baseline specification, quantifying the direct effects of environmental, market, and product attributes on alfalfa hay prices. The general form of the regression model is represented as:

$$Price_{it} = \beta_0 + \beta_1 DSCI_{it} + \beta_2 Precip_{it} + \beta_3 Temp_{it} + \beta_4 Corn_{it} + \beta_5 Feeder_{it} + \beta_6 \sum_{k=1}^{K-1} Quality_{it}^k + \beta_7 \sum_{j=1}^{J-1} Package_{it}^j + \beta_8 Exp_{it}$$

Where $Price_{it}$ is Alfalfa hay price in state i at time t ; $DSCI_{it}$ is Drought Severity and Coverage Index, $Precip_{it}$ is Precipitation; $Temp_{it}$ is Temperature; $Quality_{it}^k$ is the set of $K-1$ dummy variables for forage quality categories; $Package_{it}^J$ is represents the set of $J-1$ dummy variables for packaging types; Exp_{it} is Export volume to China; $Corn_{it}$ is Corn price; $Feeder_{it}$ is Feeder cattle price; γ_s and δ_m are State and Month fixed effects; and ϵ_{it} is Error term.

This model establishes the foundational relationships among drought, market, and product factors, serving as a benchmark for subsequent interaction models.

(B) Trade Buffer Model

The Trade Buffer Model extends the baseline by including an interaction between drought severity and export volume:

$$Price_{it} = \beta_0 + \beta_1 DSCI_{it} + \dots + \beta_9 (DSCI_{it} \times Exp_{it}) + \gamma_s + \delta_m + \epsilon_{it}$$

This specification tests whether international trade activity moderates the effect of drought on local alfalfa prices. The model is motivated by the hypothesis that regions characterized by elevated export volumes may exhibit either reduced or enhanced price responses to drought, dependent on the moderating or enhancing influences of worldwide demand.

(C) Market Segmentation Model

The Market Segmentation Model incorporates an interaction between drought severity and packaging type:

$$Price_{it} = \beta_0 + \beta_1 DSCI_{it} + \dots + \beta_9 (DSCI_{it} \times Package_{it}) + \gamma_s + \delta_m + \epsilon_{it}$$

This model examines whether the price impact of drought varies by bale format, which serves as a proxy for market segmentation (e.g., export-oriented versus domestic channels). The rationale is that drought may affect prices differently across packaging types due to differences in market destination and buyer sensitivity.

(D) Quality Boost Model

The Quality Boost Model introduces an interaction between drought severity and forage quality:

$$Price_{it} = \beta_0 + \beta_1 DSCI_{it} + \dots + \beta_9 (DSCI_{it} \times Quality_{it}) + \gamma_s + \delta_m + \epsilon$$

This allows us to test whether the price impact of drought varies by bale format, reflecting differences in market segmentation.

(E) Inventory Moderation Model

The Inventory Moderation Model extends the baseline regression by incorporating inventory levels measured on May 1 and December 1. These inventory variables capture supply-side stock conditions that influence alfalfa price formation alongside drought severity. To isolate the combined effects of drought and inventory on prices, this model excludes precipitation and temperature variables, focusing on how carryover stocks moderate drought impacts. Additionally, interaction terms between drought severity and inventory levels are included to test whether the availability of hay stocks dampens or amplifies price responses during drought periods. This approach allows for an examination of how seasonal inventory buffers shape alfalfa market resilience in the face of environmental stress.

$$Price_{it} = \beta_0 + \beta_1 DSCI_{it} + \dots + \beta_9 Stock_{it} + \beta_{10} (DSCI_{it} \times Stock_{it}) + \gamma_s + \delta_m + \epsilon_{it}$$

Where $Stock_{it}$ represents the current alfalfa inventory in state i and month t .

All models are estimated with state and month fixed effects and standard errors clustered at the state level. The regression results are interpreted as associations rather than causal effects, but they provide important insights into how drought interacts with supply, trade, and product differentiation in shaping forage market outcomes. By structuring these models as the Drought Market Model, Trade Buffer Model, Market Segmentation Model, Quality Boost Model, and Inventory Moderation Model, a clear analytical framework is provided that highlights the novel contributions of each specification and facilitates the interpretation of the results. Together, a comprehensive empirical strategy is formed by these models to quantify both the independent and compounded effects of drought in a dynamic, multi-factor pricing environment. Although causal effects are not inferred from the regression results, valuable insights are gained regarding the interactions between drought, supply, and trade conditions in influencing forage market outcomes.

2.4 Results

This section presents the empirical findings from the analysis of alfalfa hay price formation in nine Western U.S. states (California, Idaho, Montana, Colorado, Arizona, Utah, New Mexico, Oregon, Washington) from August 2021 to September 2024. The results address the research objectives outlined in Chapter 1, specifically quantifying the impacts of drought severity, forage quality, packaging type, export volumes, feed market prices, and inventory levels on alfalfa prices, as modeled in Section 2.3. The findings are organized into three

subsections: descriptive statistics, visual analysis of key trends, and regression results, followed by a discussion of limitations.

2.4.1 Descriptive Statistics and Trends

Table 3 summarizes the key variables used in the analysis. Alfalfa hay prices averaged \$279.58 per ton, with a wide range (\$21–\$540), reflecting significant market volatility. Drought severity, measured by the Drought Severity and Coverage Index (DSCI), averaged 179.98 (SD = 123.85), with peaks reaching 431.2 during extreme drought episodes. Export volumes to China averaged 112.52 thousand metric tons monthly, while corn and feeder cattle prices showed notable variability, underscoring their role as demand drivers. Inventory levels, precipitation, and temperature further highlight the climatic and supply-side diversity across the study region.

Table 3.
Summary Statistics of Key Variables

Variable	mean	Std Dev	Median	Min	Max	Std Err
Alfalfa Price(\$/ton)	279.58	91.97	270	21	540	1.14
Drought Severity	179.98	123.85	160.75	1.5	431.2	1.53
Precipitation	44.42	41.19	31.39	0.335	246.43	0.51
Temperature	15.28	8.16	16.65	-9.96	29.13	0.10
Export Volume	112.52	50.44	102.11	46.49	240.48	0.62
Inventory level (1,000tons)	621.23	609.86	325	30	2550	23.23
Corn Price(\$/Bu)	5.57	1.19	5.49	3.77	7.89	0.01
Feeder Price (\$/cwt)	206.27	38.85	190.84	155.62	258.28	0.48

Figure 6 presents a plot of average alfalfa prices across the nine states from August 2021 to September 2024. Prices rose sharply from \$250/ton in late 2021 to over \$350/ton in mid-2022, coinciding with severe drought conditions, before stabilizing around \$280/ton by mid-2024. Figure 7 further highlights the substantial state-level variation in alfalfa hay prices from 2021 to 2024. California, Colorado, and New Mexico exhibit the highest median prices and the widest price ranges, indicating both strong demand and greater price volatility, likely due to their large dairy sectors and significant export activity as mentioned in literature. Utah and Idaho, while important producers, display lower median prices and less dispersion, suggesting more stable but less lucrative market conditions. The presence of outliers and wide price ranges in several states suggests the influence of factors such as drought, supply shortages, or fluctuating export demand, though the exact causes are not always clear. Collectively, these patterns reflect a highly segmented regional market in which both intrinsic product characteristics (quality, packaging) and extrinsic factors (state-level supply-demand dynamics, drought impacts, and export flows) interact to drive alfalfa price variability across the West.

Figure 6.
Alfalfa Average Price Over Time (2021 – 2024)

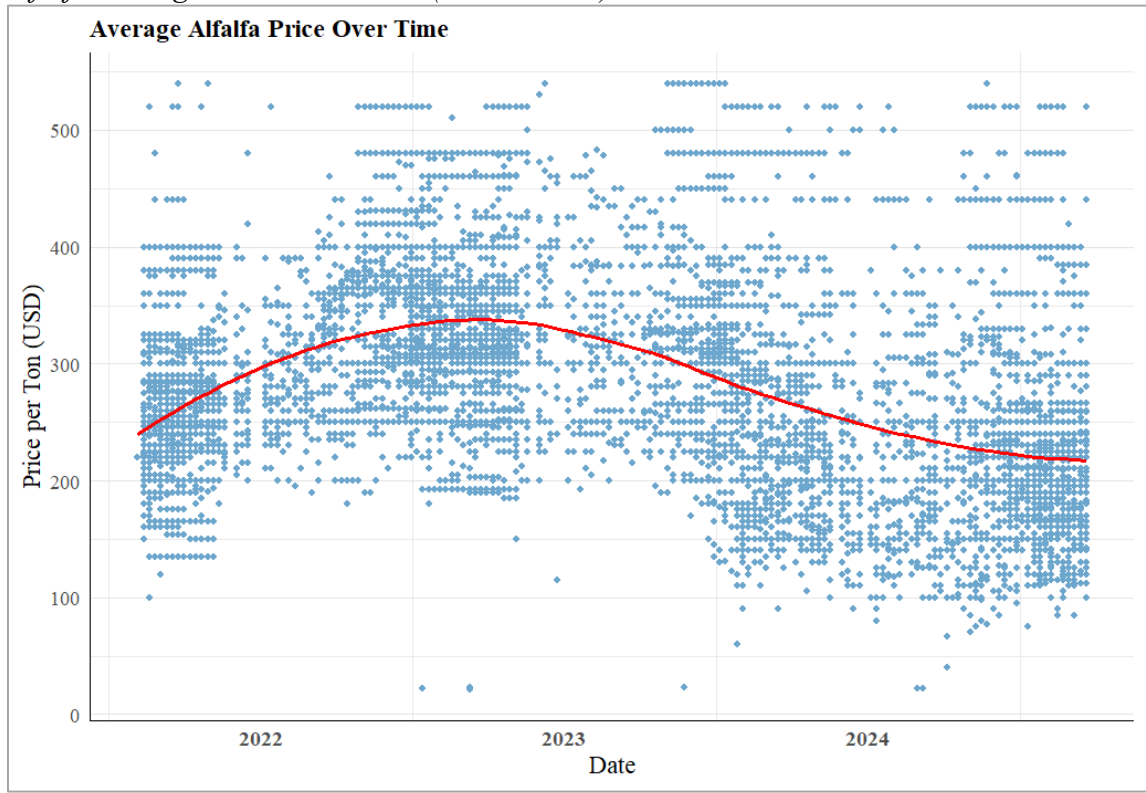
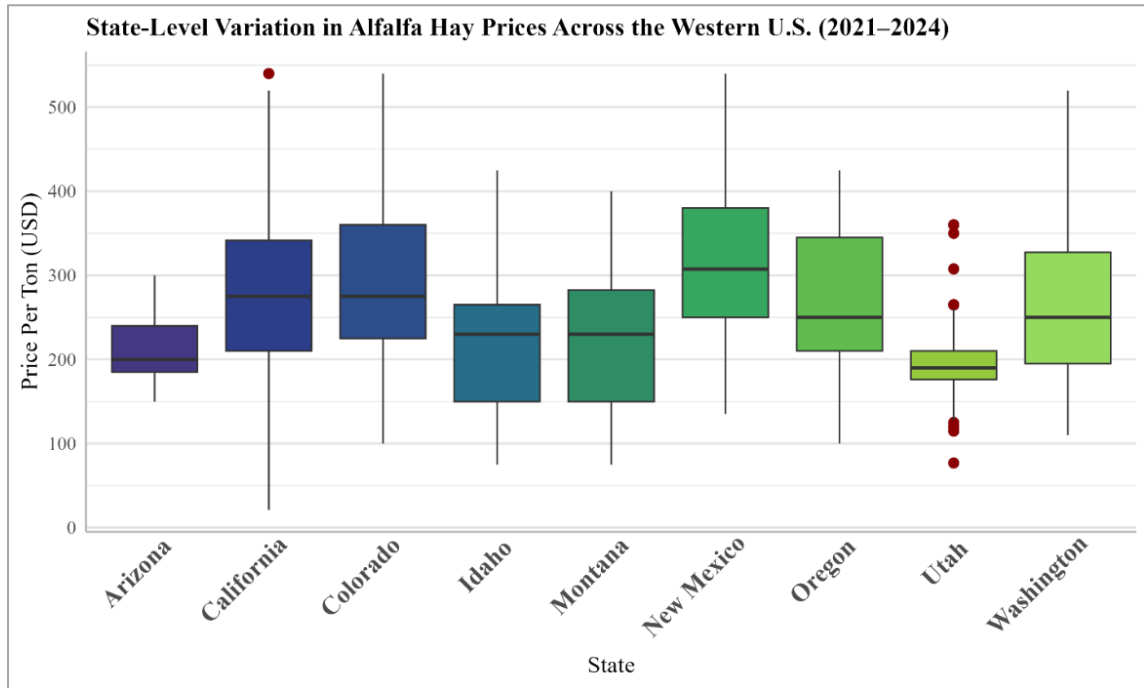
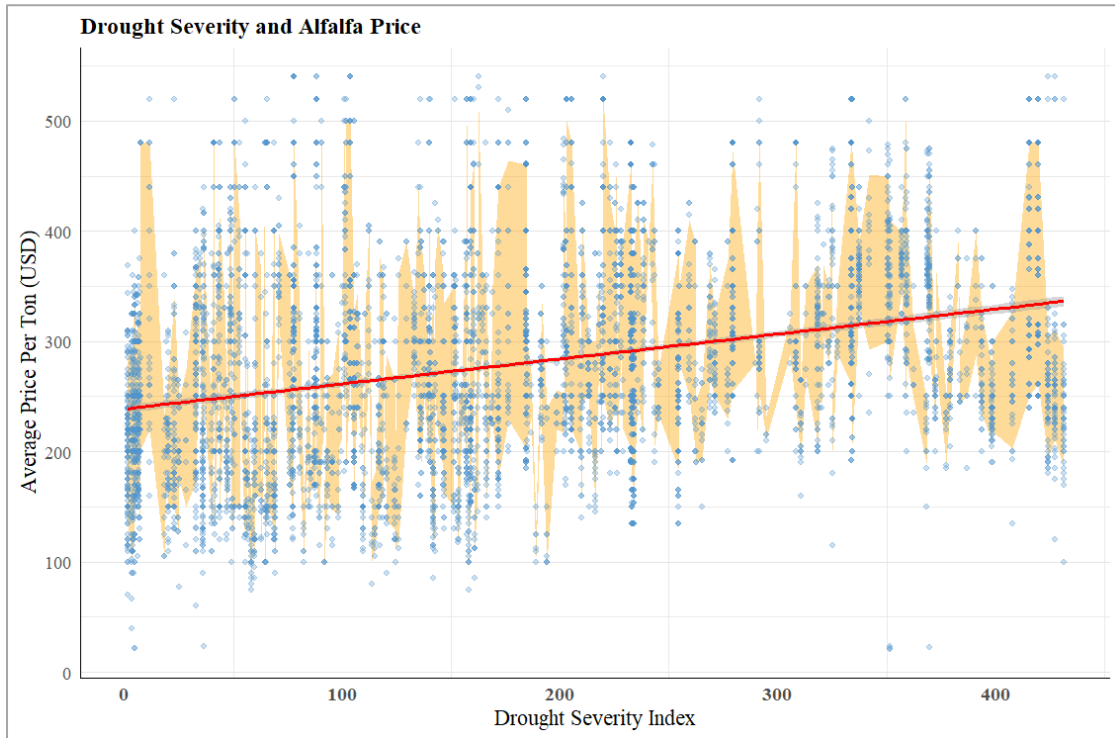


Figure 7.*Alfalfa Hay Prices (USD/Ton) Across Western U.S. States (2021 To 2024)*

Note. Boxplot highlights Alfalfa regional price variation across western states.

Figure 8 further explores the drought-price relationship. When drought severity is plotted directly against alfalfa price, a positive correlation becomes evident. Periods of higher drought severity are associated with elevated prices. This alignment underscores the sensitivity of forage markets to environmental stress, where drought conditions can tighten supply and push prices upward, amplifying volatility in the regional hay economy.

Figure 8.
Alfalfa Price and Drought Severity Relationship (2021-2024)

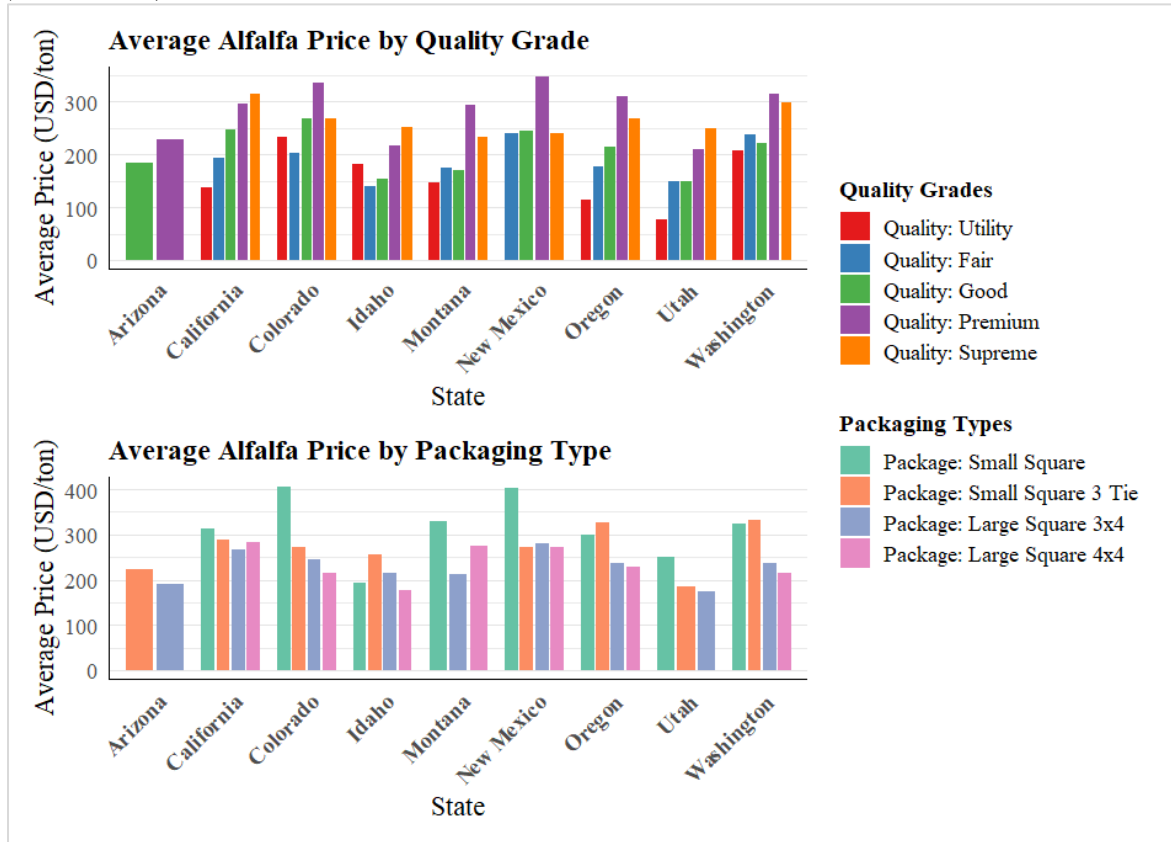


2.4.2 Role of Product Characteristics

As shown in Figure 9, average alfalfa prices differ significantly by packaging type and quality grade. Small Square Bales consistently command the highest prices across all states, often exceeding \$400 per ton, reflecting their appeal in retail markets where ease of handling and consistent quality are important. In comparison, Large Square Bales (3x4 and 4x4) are priced lower, serving bulk buyers and export demand. Quality grade also plays a major role. Supreme and premium alfalfa typically receive price premiums of \$100 to \$150 per ton above fair or utility grades, highlighting the market's sensitivity to forage quality factors like relative feed value and digestibility.

Figure 9.

Average Alfalfa Hay Packaging and Quality Impact on Price Across Western U.S. States (2021–2024)



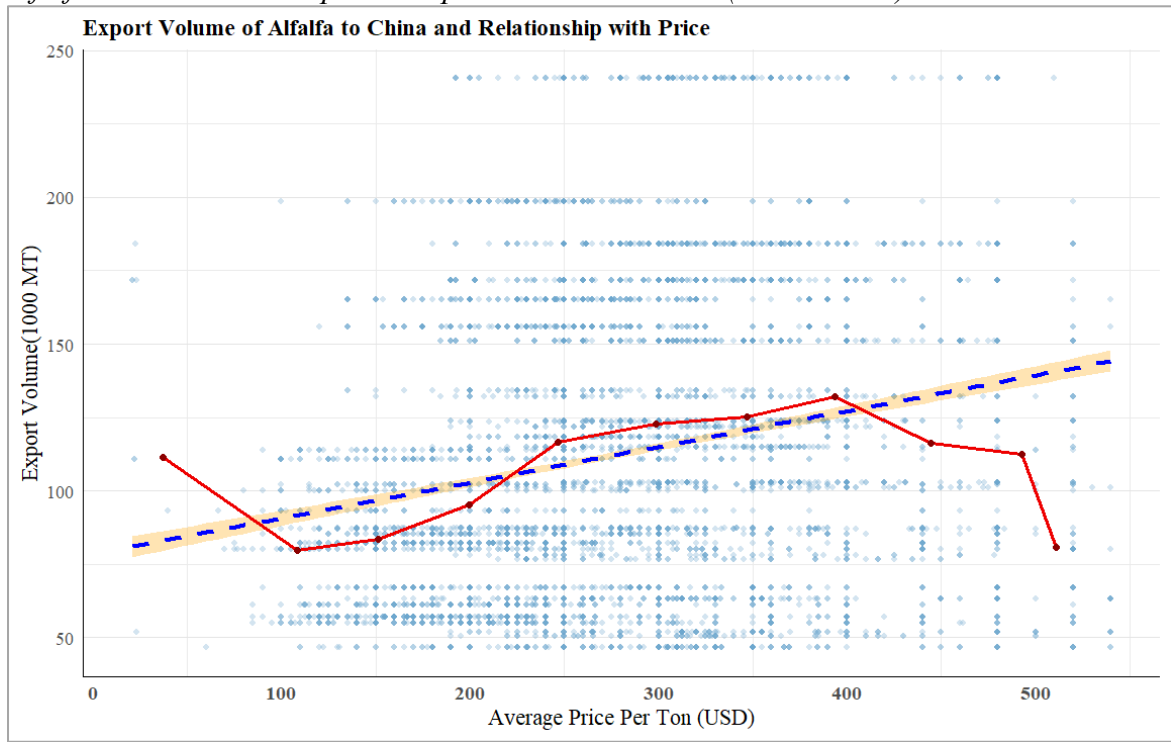
2.4.3 Influence of Export Activity and Feed Market Prices

Export activity, particularly shipments to Asian markets, is positively correlated with alfalfa price formation. Figure 10 presents the relationship between the average alfalfa price per ton and monthly export quantity to China. The scatterplot reveals a generally positive relationship between average prices and export quantities, with exports increasing as prices rise up to a certain point. The fitted trend line and binned averages indicate that across most of the price range, higher prices are associated with greater export volumes, demonstrating strong international demand even when prices are elevated. However, at the highest price

levels, export quantities decline, indicating that international buyers, specifically China, may become more price-sensitive or that supply constraints limit exports when prices are exceptionally high. This pattern highlights the dual role of export markets in both supporting prices and imposing an upper limit on demand during extreme market conditions.

Figure 10.

Alfalfa Price Relationship with Export Volume to China (2021 – 2024)

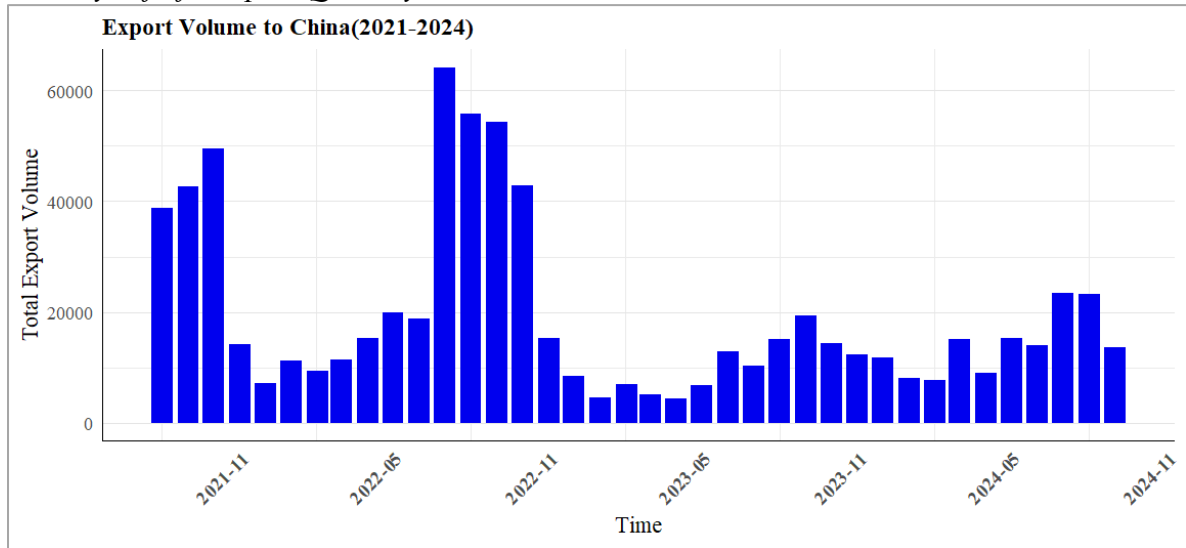


In addition, Figure 11 illustrates the monthly pattern of alfalfa export quantities from 2021 to 2024. Export activity exhibits pronounced seasonality, with shipments peaking during the summer and early fall months, which coincides with the main harvest and marketing periods. The year 2022 stands out, with exceptionally high export volumes, particularly in mid-year, aligning with the severe drought and elevated prices observed in the Western U.S. In contrast, export quantities in 2023 and 2024 are lower and more evenly distributed across months, reflecting a return to more typical market conditions as drought pressures eased.

These trends underscore the responsiveness of export markets to both seasonal production cycles and broader market shocks, such as drought-driven supply disruptions.

Figure 11.

Monthly Alfalfa Export Quantity to China From 2021 To 2024



2.4.4 Baseline Regression Model Results

Table 4 presents the results from the baseline fixed-effects regression model, which estimates the associations between alfalfa hay prices and a range of environmental, market, and product factors across Western U.S. states. The model includes state and month fixed effects, and standard errors are clustered at the state level to account for within-state correlation over time. This approach helps isolate the relationships between drought, export activity, market drivers, and product attributes with alfalfa prices, net of persistent spatial and temporal confounders. The results indicate that the Drought Severity Index (DSI) shows a positive coefficient (0.087), suggesting a modest association with higher prices, though it is not statistically significant. Specifically, a one-unit increase in the Drought Severity Index is associated with a \$0.087 per ton increase in price. Other environmental

variables include precipitation and temperature are not statistically significant. This lack of significance for precipitation is consistent with the predominance of irrigated alfalfa production in the Western U.S., where direct rainfall plays a limited role in yield determination. Forage quality significantly correlates with price variations. The combined alfalfa categories (such as Good/Premium or Utility/Fair) commonly used in USDA market reports reflect an overlap in relative feed value (RFV), where hay quality characteristics fall between defined grading thresholds or when market lots average across both categories due to variability in forage testing and visual assessment. Lower-quality categories like Utility (−\$95.28) and Fair (−\$71.76) associated with substantial price reductions compared to the Premium baseline, while Supreme quality shows a positive but non-significant coefficient. This suggests limited variation in prices among top-tier quality grades, potentially due to buyer preferences or market saturation for premium hay

Packaging type also matters, with Small Square bales commanding a significant premium (\$100.42) over Large Square 4x4 bales (−\$21.45). Corn prices (\$46.54 per bushel) and feeder cattle prices (\$0.73 per cwt) exhibit strong positive correlations with alfalfa prices, reflecting feed market linkages. Among market factors, corn prices have a strong and significant correlation, and a one dollar increase in corn price is associated with a \$34.01 per ton increase in alfalfa hay price. In this model, feeder cattle price and export quantity do not have statistically significant effects. Export quantity shows no significant associations ($p > 0.1$). The model explains 57% of within-state variation (Within $R^2 = 0.57$) and 64% of total variation (Adjusted $R^2 = 0.64$), with a root mean square error of 55.2, indicating a robust fit for analyzing price dynamics under drought and market conditions.

Table 4.
Baseline Model Regression Results for Alfalfa Hay Prices, (2021-2024)

Variable	Coefficient	Std. Error	P-value	Significance
Drought Severity Index (DSCI)	0.087	0.056	0.159	
Quality				
Fair	-71.756	4.891	0.000	***
Fair/Good	-60.154	11.657	0.001	***
Good	-51.046	4.657	0.000	***
Good/Premium	-26.849	3.711	0.000	***
Premium/Supreme	3.642	6.560	0.594	
Supreme	12.804	7.244	0.115	
Utility	-95.279	11.399	0.000	***
Utility/Fair	-68.875	3.965	0.000	***
Package				
Large Square 4x4	-21.451	7.129	0.017	**
Small Square	100.420	20.117	0.001	***
Small Square 3-Tie	33.168	12.919	0.033	**
Export Quantity (1000MT)	0.026	0.089	0.779	
Corn Price(\$/Bu)	46.542	4.789	0.000	***
Feeder Cattle Price(\$/cwt)	0.725	0.256	0.022	**
Precipitation(mm)	0.013	0.068	0.851	
Temperature(°C)	0.481	0.670	0.494	
Within R ² : 0.57	Fixed Effects: State and Month			
Adjusted R ² : 0.64	Standard Errors: Clustered at the state level			
RMSE: 55.2	Significance codes:			
	*** p < 0.01, ** p < 0.05, * p < 0.1			

2.4.5 *Interaction Model Results*

The interaction models provide deeper insight into how drought severity interacts with forage quality, packaging, and export activity to influence alfalfa hay prices across the Western U.S. states. Results indicate that the effect of drought on price is not uniform but is significantly shaped by product and market characteristics. A summary comparison of key coefficients from these models is provided in Table 5.

A key observation is that, in the trade buffer model, drought severity (DSCI) is positively and significantly correlated with alfalfa hay prices. Specifically, a one-unit increase in DSCI is associated with a \$0.16 per ton increase in price ($p = 0.022$), holding all other factors constant. This suggests that periods of more severe drought tend to coincide with higher alfalfa prices in the Western U.S., likely reflecting tighter forage markets and increased competition for available supply. It is important to note, however, that this relationship is correlational rather than strictly causal, as other unobserved factors or market dynamics may also contribute to the observed association. Export quantity and the drought $\times$ export interaction does not show statistically significant relationships with price in this model. The inclusion of these terms does not alter the lack of direct association between export activity and price, nor does it indicate that exports buffer or amplify the impact of drought on prices under current conditions. However, their inclusion helps clarify the unique association between drought and price by reducing potential confounding from omitted variables, which may have masked the drought relationship in simpler model specifications.

Other covariates remain consistent with baseline findings. Forage quality and packaging type are major determinants of price. Higher quality grades such as Good, Good/Premium, Premium, Premium/Supreme, and Supreme command substantial and statistically significant premiums, while Utility grade is associated with a significant discount. Packaging effects are strong, with Small Square Bales and Small Square 3-Tie Bales receiving significant premiums, and Large Square 4x4 packaging associated with a significant discount. Among market factors, corn price again has a strong and significant correlation with alfalfa price, and feeder cattle price also shows a positive and significant association. Precipitation and temperature remain statistically insignificant, which aligns with the predominance of irrigated production in the Western U.S., where direct weather effects are less pronounced. In summary, the trade buffer model indicates a strong positive correlation between drought severity and alfalfa hay prices, even after accounting for export activity and other market factors. The emergence of this relationship in the buffer model, compared to its absence in the baseline model, underscores how model specification and the inclusion of relevant variables can clarify the nature of observed associations. Compared to the baseline regression, the trade buffer model maintains similar explanatory power, with an identical within R^2 of 0.57 and adjusted R^2 of 0.64, but achieves a slightly improved fit, as evidenced by a marginally lower root mean squared error (RMSE) of 55.1 compared to 55.2 in the baseline model.

The market segmentation model investigates how drought severity affects alfalfa hay prices across different packaging types. The findings show that drought severity (DSCI) has a significant and positive influence on alfalfa hay prices in this model. Specifically, a one-unit increase in DSCI is associated with a \$0.14 per ton rise in price ($p = 0.035$), holding

other factors constant. This result highlights the ongoing upward pressure that drought places on forage markets due to supply constraints and increased competition during dry periods.

Crucially, the interaction terms between drought severity and packaging types reveal important market segmentation effects. The interaction between drought and Large Square 4x4 packaging is negative and statistically significant, as is the interaction with Small Square Bales. The interaction with Small Square Bales3-Tie packaging is also negative and marginally significant. These results indicate that during periods of severe drought, the typical price premiums for these specialty packaging formats are reduced relative to Large Square 3x4. In other words, as drought severity increases, buyers are less willing to pay extra for Large Square 4x4 and Small Square Bales, suggesting that market preferences shift toward basic forage availability rather than packaging format. Higher quality grades continue to command significant premiums relative to the reference category, with price increases ranging from \$13.03 per ton for Supreme up to \$1.88 per ton for Premium/Supreme (not statistically significant in this model). Small Square Bales command a substantial premium of \$148.5 per ton, and Small Square 3-Tie Bales receive a premium of \$46.67 per ton compared to Large Square 3x4. Large Square 4x4 packaging does not show a significant main effect relative to the reference.

Among market factors, corn price is again a strong and significant predictor, with a one dollar increase in corn price associated with a \$46.30 per ton increase in alfalfa price. Feeder cattle price is also positively and significantly associated with alfalfa price. Precipitation, temperature, and export quantity do not exhibit statistically significant effects in this model. These findings demonstrate that drought not only raises overall alfalfa

hay prices but also compresses the price premiums for certain packaging types, particularly Large Square 4×4 and Small Square Bales. This suggests that during drought, buyers may become less willing to pay extra for specialty packaging, possibly due to tighter supply or shifting priorities toward availability over packaging format.

Quality boost model incorporates interactions between drought severity and forage quality, provides further insight into how environmental stress shapes the price structure of alfalfa hay in the Western United States. The model indicates that the direct effect of drought severity (DSCI) on Premium-quality alfalfa hay prices is not statistically significant ($p = 0.502$), suggesting that drought alone does not directly alter the price for the highest-quality hay. However, the interaction terms between drought and forage quality reveal a substantial shift in how drought influences price formation across different quality grades. By including $DSCI \times \text{Quality}$ interactions, lower-quality grades like Fair (\$0.198 per ton) and Utility (\$0.190 per ton) show price increase under severe drought, while Premium/Supreme shows a price reduction (−\$0.110 per ton). It reveals that drought compresses the price spread between high- and low-quality hay. During periods of heightened drought, buyers are less able to pay large premiums for top-quality forage, likely due to overall scarcity and the need to secure any available supply. As a result, the traditional quality premium narrows, and the market becomes less segmented by quality. Forage quality remains a major determinant of price with all lower grades are associated with significant discounts relative to Premium, and only the Premium/Supreme category commands a significant positive premium. Packaging effects are also strong, with Small Square Bales commanding a \$99.69/ton premium and Small Square 3-Tie Bales \$33.12/ton, while Large Square 4x4 packaging is associated with a \$21.56/ton discount. Other market factors remain

significant. Corn price has a strong and significant correlation, with a one dollar increase in corn price associated with a \$47.38 per ton increase in alfalfa price. Feeder cattle price is also positively and significantly associated with alfalfa price. Precipitation, temperature, export quantity, and do not exhibit statistically significant effects in this model. In summary, drought no longer raises all alfalfa prices evenly. This shows that it's important to look at how drought and quality work together to understand how the market reacts during dry periods.

The inventory moderation model explores whether inventory levels buffer drought-driven price volatility. This approach tests the hypothesis that higher available stocks moderate price pressures. In this specification, precipitation and temperature were excluded as control variables to focus on the direct effects of drought severity. This adjustment ensures that the estimated effect of drought reflects its composite influence, as captured by the DSCI, without potential redundancy from including individual weather variables. Results indicate that the coefficient for the current inventory is negative. However, this effect is not statistically significant, indicating that, during the study period, inventory levels did not have an effective or direct influence on alfalfa hay prices after accounting for other key factors. In contrast, drought severity (DSCI) displays a positive and marginally significant association with price, and corn price emerges as a strong, significant predictor. Forage quality and packaging variables also show expected patterns with lower quality grades with significant price discounts, while higher quality grades and certain packaging types, such as Small Square Bales, command notable premiums. The regression results show a significant negative association between export quantity and alfalfa hay prices. The negative export coefficient is significant only in the inventory model, suggesting that high

export volumes might lower prices when inventory is controlled and weather variables are excluded. This suggests that higher export volumes are linked to lower domestic prices during the study period, potentially reflecting the influence of international demand dynamics and the role of exports in absorbing excess supply or shifting market equilibrium. These finding highlights that, alongside drought and feed market conditions, export activity remains a key factor influencing price volatility in the Western U.S. alfalfa market. Model diagnostics indicate improved fit over the baseline ($R^2 = 0.57$, RMSE = 55.2), with the Inventory Model achieving the highest adjusted R^2 (0.706) and the Quality Boost Model the lowest RMSE (54.05), underscoring the value of incorporating interaction terms to capture complex price dynamics under drought conditions. The Inventory Model exhibits the highest explanatory power among the interaction models, with an adjusted R^2 of 0.706, indicating a strong fit to the data. However, with limited observations due to available data just for May and December, this high R^2 may reflect potential overfitting or bias, particularly if the sample is not fully representative of broader market dynamics.

Table 5.
Interaction Models for Alfalfa Price Determination in The Western U.S.

Variable	Models			
	Trade Buffer	Market	Quality Boost	Inventory
Drought Severity Index (DSCI)	0.160** (0.057)	0.143** (0.056)	0.049 (0.069)	0.197* (0.090)
Export Quantity	0.150 (0.203)	0.045 (0.086)	0.056 (0.077)	-1.187*** (0.235)
DSCI $\times$ Export Quantity	-0.001 (0.001)			

Variable	Models			
	Trade Buffer	Market	Quality Boost	Inventory
Quality				
Fair	-71.624*** (5.058)	-71.709*** (5.666)	-103.942*** (8.480)	-88.259*** (9.887)
Fair/Good	-59.606*** (11.707)	-59.410*** (11.156)	-81.865*** (13.722)	-73.207*** (14.852)
Good	-50.942*** (4.703)	-50.010*** (4.811)	-62.645*** (7.174)	-51.323** (15.059)
Good/Premium	-26.160*** (3.193)	-27.254*** (3.671)	-39.111** (13.459)	-9.352 (10.716)
Premium/Supreme	4.440 (6.082)	1.877 (4.977)	28.670** (11.076)	4.259 (15.727)
Supreme	12.907 (7.169)	13.026* (6.458)	1.983 (12.795)	17.645* (8.400)
Utility	-94.982*** (11.631)	-96.932*** (10.245)	-122.148*** (16.659)	-105.060*** (23.051)
Utility/Fair	-68.583*** (4.128)	-67.579*** (3.813)	-104.523*** (9.830)	-72.658*** (18.674)
DSCI × Quality: Fair			0.198*** (0.037)	
DSCI × Quality: Fair/Good			0.136 (0.086)	
DSCI × Quality: Good			0.071** (0.030)	
DSCI × Quality: Good/Premium			0.069 (0.088)	
DSCI × Quality: Premium/Supreme			-0.110* (0.051)	

Variable	Models			
	Trade Buffer	Market	Quality Boost	Inventory
DSCI × Quality: Supreme			0.065 (0.036)	
DSCI × Quality: Utility			0.190*** (0.043)	
DSCI × Quality: Utility/Fair			0.225*** (0.051)	
Package				
Large Square 4x4	-20.876** (7.000)	6.871 (16.139)	-21.556** (7.042)	-36.848* (19.371)
Small Square	100.571*** (20.083)	148.542*** (26.203)	99.690*** (19.346)	110.974*** (23.064)
Small Square 3-Tie	33.057** (12.870)	46.665*** (13.794)	33.115** (12.523)	34.314* (14.805)
DSCI × Package: Large Square 4x4		-0.139** (0.043)		
DSCI × Package: Small Square		-0.231** (0.087)		
DSCI × Package: Small Square 3Tie		-0.084* (0.044)		
Precipitation(mm)	0.019 (0.063)	0.006 (0.069)	0.029 (0.066)	
Temperature(°C)	0.192 (0.456)	0.558 (0.764)	1.001 (0.744)	
Corn Price(\$/Bu)	45.745*** (4.889)	46.296*** (4.305)	47.382*** (4.942)	61.290** (20.135)
Feeder Price(\$/cwt)	0.721**	0.717**	0.760**	1.455

Variable	Models			
	Trade Buffer	Market	Quality Boost	Inventory
	(0.262)	(0.247)	(0.243)	(0.868)
Inventory Level (1000 tons)				-0.002 (0.015)
R ²	0.639	0.650	0.653	0.715
R ² Adj.	0.637	0.647	0.651	0.706
RMSE	55.13	54.32	54.05	54.64
Fixed Effects: State and Month				
Standard Errors: State level				
Significance codes: * p < 0.1, ** p < 0.05, *** p < 0.01				

2.5 Discussion

Our results demonstrate that drought compresses the price premiums typically associated with higher forage quality and preferred packaging formats, indicating that during periods of environmental stress, buyers prioritize availability over specific product attributes. The negative but insignificant coefficient for inventory suggests that other factors, particularly drought conditions, feed market prices, and product characteristics, have played a more dominant role in shaping price volatility. Similarly, export activity, measured by shipments to China, is significantly negatively associated with price, indicating that higher export volumes are linked to lower domestic prices, likely reflecting the influence of international demand dynamics. However, export activity did not significantly moderate the relationship between drought and price during the study period. Weather variables such as precipitation and temperature generally exhibit weak direct associations with hay prices, with their effects largely captured by the composite drought index. These results highlight the complexity of

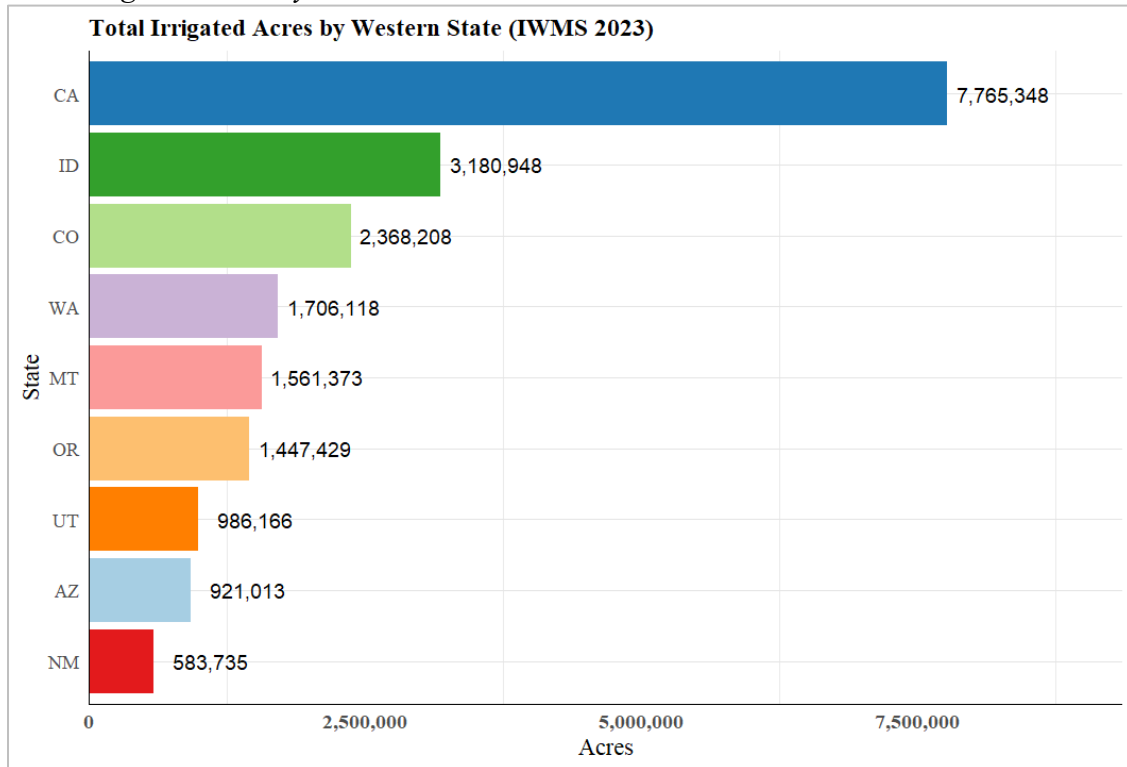
alfalfa price formation and reinforce the need for producers, exporters, and policymakers to consider a broad set of market, environmental, and product factors when making decisions under increasing climate and trade uncertainty.

While the analysis offers important insights, several limitations should be acknowledged. Potential endogeneity between supply and price such as reverse causality (where higher prices may incentivize inventory drawdown or affect reporting) or omitted variable bias cannot be fully ruled out. Although the use of fixed effects mitigates some confounding, unobserved factors and simultaneity may still influence the estimated relationships. Measurement error in forage quality and packaging variables, as well as reliance on projected or preliminary data for late 2024, may introduce additional uncertainty. Unobserved confounding factors and potential endogeneity between supply and price cannot be fully ruled out. Overall, our findings underscore the importance of product differentiation, market integration, and adaptive management strategies for navigating the challenges of a changing climate and volatile market conditions in the Western U.S. alfalfa sector

CHAPTER 3: MODERNIZING WHEEL-LINE IRRIGATION SYSTEMS: A PATH TO SUSTAINABLE ALFALFA PRODUCTION

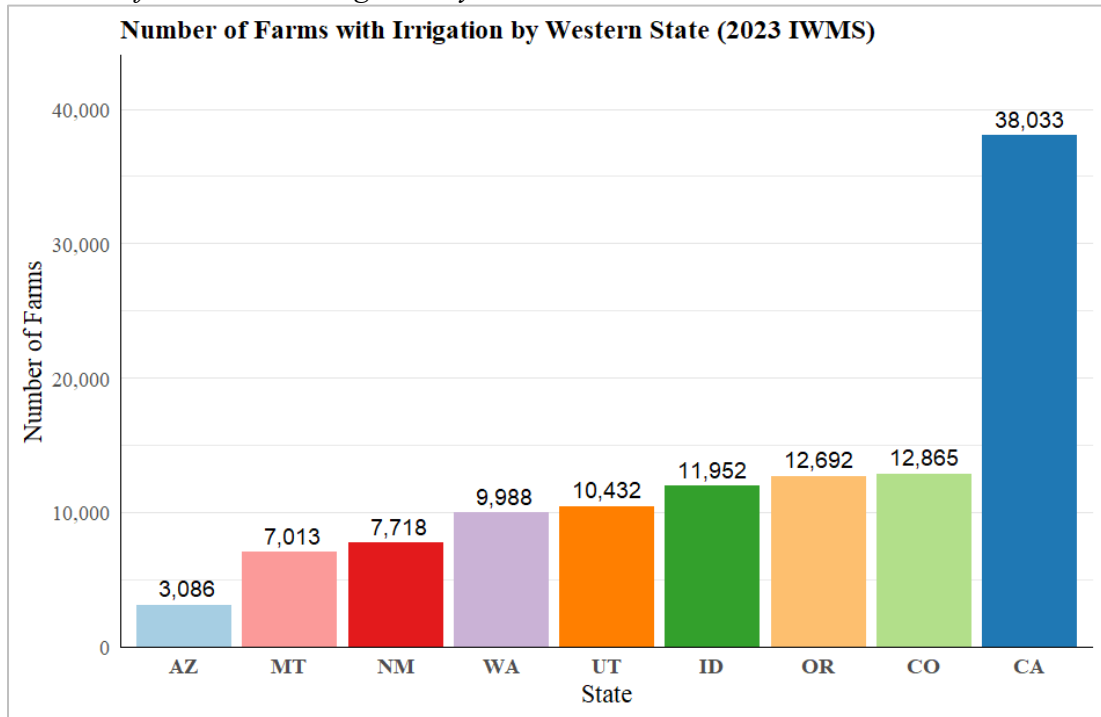
3.1 Introduction

Irrigation serves as the lifeblood of agricultural production across the Western United States, with California maintaining its position as the dominant irrigated state with more than 7.7 million irrigated acres according to the 2023 Irrigation and Water Management Survey (IWMS) and it is followed by Idaho, Colorado, and Washington, which collectively account for nearly 70% of the region's irrigated farmland (*IWMS*, 2023). The critical importance of irrigation is further underscored by the number of farming operations dependent on these systems, with California alone hosting more than 38,000 irrigated farms and significant concentrations present in Oregon, Colorado, and Idaho (*IWMS*, 2023). Figures 12 and 13 demonstrate the fundamental role of irrigation in sustaining Western agricultural systems.

Figure 12.*Total Irrigated Acres by Western State*

Note. California leads irrigated acres in western states with over 7.7 million acres, followed by Idaho, Colorado, and Washington based on 2023 Irrigation and Water Management Survey (IWMS, 2023)

Figure 13.
Number Of Farms with Irrigation by Western State



Note. California has the highest count at 38,033 farms, with significant numbers also observed in Oregon, Colorado, and Idaho according to 2023 Irrigation and Water Management Survey (IWMS, 2023)

Within this context, wheel-line irrigation systems have maintained their prominence as a common technology for alfalfa production. Wheel-line irrigation systems consist of large wheels supporting a central pipeline that delivers water to sprinklers and is periodically moved across the field to ensure even irrigation coverage (Waller & Yitayew, 2016). These systems are preferred by many farmers due to their relatively low capital cost, adaptability to irregular or sloped fields, and effectiveness for irrigating a variety of forage crops such as alfalfa (Milhollin, 2017; Sanden et al., 2011). As shown in Figure 14, California, Idaho, and Oregon account for the largest areas irrigated via wheel-line systems, reflecting both historical investment patterns and ongoing practical considerations in these states (IWMS, 2023). When examining farm operations rather than total acreage, Idaho and Montana

emerge as leaders in wheel-line adoption, highlighting the continued relevance of this technology among small- to medium-scale producers as shown in Figure 15 (IWMS, 2023).

Figure 14.

Total Acres Irrigated Using Wheel Line Systems by State.

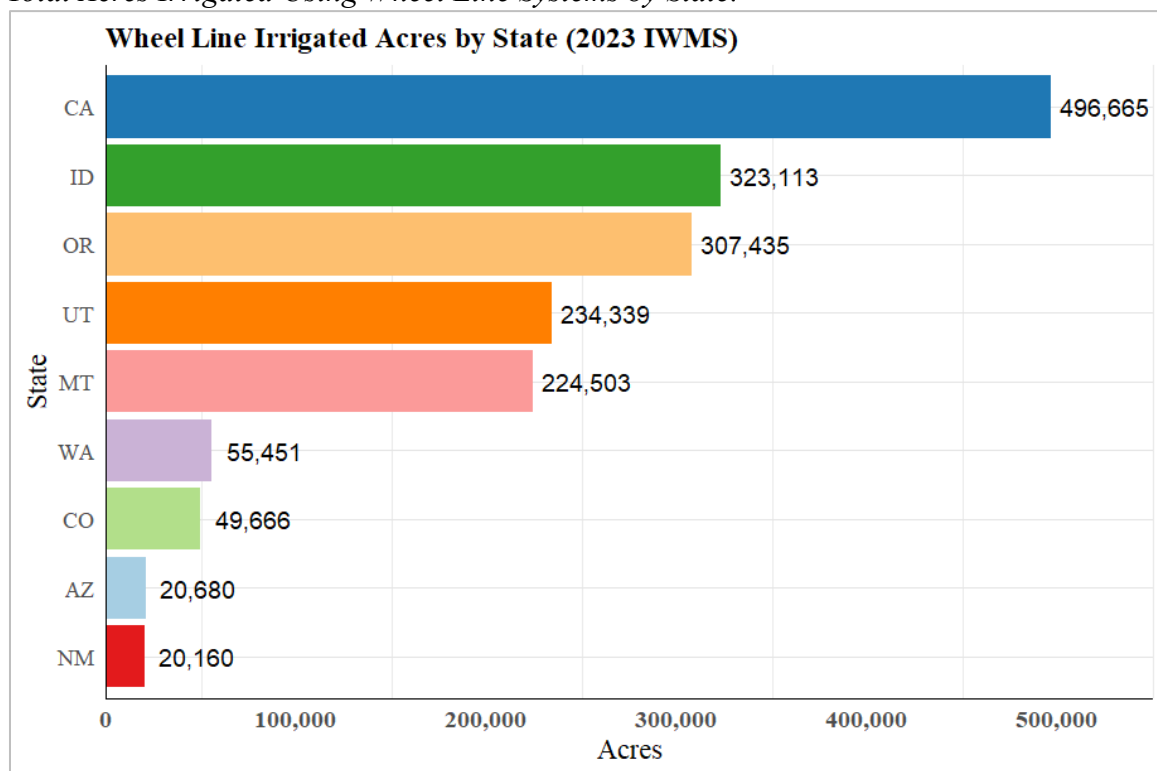
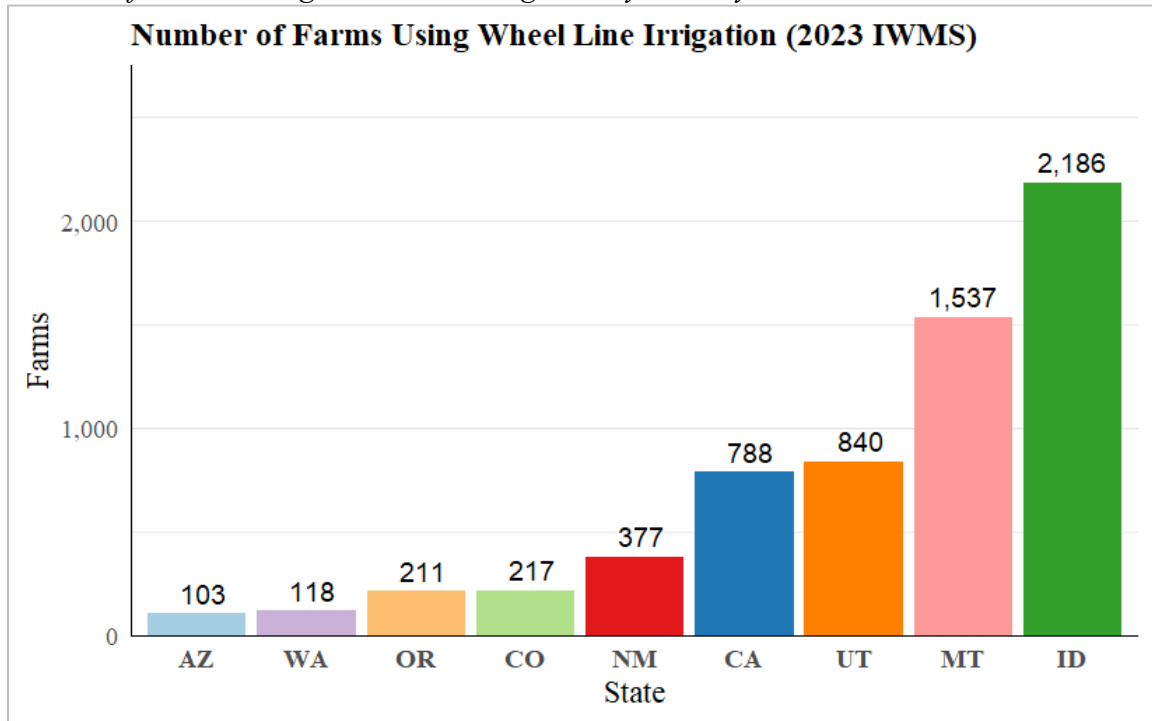


Figure 15.*Number of Farms Using Wheel Line Irrigation Systems by State*

Despite their reliability and low cost, wheel-line systems are much more labor-intensive than automated methods like center pivot or drip irrigation, requiring frequent manual moves and long irrigation sets. They also have lower application efficiency (45–70%) compared to pivots (75–85%) and drip systems (85–90%) (Heaton, 2008). Wheel-line systems have undergone little structural or technological innovation since their introduction, with designs remaining largely unchanged and manual operation still common (J. Wang et al., 2015). As water scarcity and energy costs rise, and the need to enhance water-use efficiency becomes more urgent, there is increasing pressure to implement cost-effective improvements to these systems (Fan & McCann, 2020; Su & Singh, 2024). Given their extensive use and continued reliance on manual operation, wheel-line irrigation systems represent a key target for modernization. Modernization opportunities for wheel-line irrigation systems, such as adoption of lower-pressure sprinkler technologies to reduce

energy consumption, integration of solar-powered, remote-controlled movers to decrease labor demands, and installation of automated shutoff valves to enhance operational efficiency, have the potential to significantly reduce labor input while improving water-use efficiency (Ikechiamaka et al., 2024; Robert W. Coates & Michael J. Delwiche, 2006). These technological advancements offer potential solutions to the systemic challenges facing traditional wheel-line systems while maintaining the economic accessibility that has made them so widely adopted. As outlined in Chapter 1, this chapter focuses specifically on the economic considerations for upgrading wheel-line irrigation systems. The primary objective is to develop and evaluate an economic framework for modernization that assesses potential gains in water-use efficiency, labor savings, and economic returns. By examining both the economic implications of these upgrades, this analysis seeks to provide actionable insights for producers facing increasing pressure to adapt their operations to changing environmental and economic conditions.

3.2 Limitations and Inefficiencies of Wheel-Line Irrigation Systems

The persistence of wheel-line irrigation systems in Western U.S. agriculture reflects a complex interplay of historical development, economic practicality, and agricultural necessity. Regional adoption patterns show that California, Idaho, and Oregon account for some of the largest acreages of alfalfa under irrigation (Orellana et al., 2025). This distribution shows the historical expansion of alfalfa production in these states during the late 20th century, when wheel-line systems emerged as the technology of choice for medium-scale operations due to their topographic adaptability and lower capital requirements (Putnam et al., 2000). However, with climate uncertainty and resource scarcity, these systems face sustainability challenges that threaten their continued viability.

The limitations of conventional wheel-line systems manifest in three critical dimensions, including water efficiency, energy consumption, and labor requirements. Field studies demonstrate application efficiencies of just 65-72% compared to 80-88% for modern alternatives. These losses are largely attributed to evaporation and wind drift, which prevent uniform water delivery and reduce effective irrigation (Faryabi et al., 2020; Stambouli et al., 2014). Energy demands compound these inefficiencies, with conventional brass-impact sprinklers requiring 50-60 psi operating pressures that results in 35-45% higher energy costs than lower pressure systems like pivots (Hanson et al., 1983; USDA NRCS, 2023).

Based on recent field data, the average energy consumption for a wheel line system is approximately 476 kWh per acre per year, which, at the 2025 average U.S. western states electricity rate of \$0.07 per kWh (*EIA*, 2025a; USDA NRCS, 2021), results in an annual energy cost of about \$33 per acre. Field studies have demonstrated that low-pressure sprinkler systems can significantly reduce energy use in maize irrigation without compromising yield (Robles et al., 2017). EIA's Short-Term Energy Outlook projects approximately 7% rise in wholesale electricity prices for 2025 compared to 2024. If irrigation energy costs follow this trend, growers may face a 10–15% increase in irrigation-related expenses over the next 5–7 years, depending on system design and usage patterns (*EIA*, 2025b).

Labor intensity presents another fundamental constraint. While precise labor-hour estimates vary by region and management intensity, extension sources consistently report that wheel-line systems demand substantially more hands-on time per acre (Scherer, 2022).. Estimates from USDA and land-grant university sources suggest that farmers spend 2–4 hours per set manually moving wheel lines every 12–24 hours, resulting in 10–15 labor hours per acre

per season and annual labor costs of \$184 to \$277 per acre, depending on field size, irrigation frequency, and local wages (McGregor, 2023; USU Extension, 2006). Over a typical growing season with 10–20 irrigation events, this results in 2.5 to 8 labor hours per acre per year (Patterson et al., 1996; USDA NRCS, 2023). Utah State University Extension reports similar figures, estimating 1.3 to 2.7 hours per acre per year, depending on crop and water management practice (USU Extension, 2006). At the current farm labor wage of \$18.43 per hour, this labor burden has become increasingly problematic considering recent workforce trends (USDA NASS, 2025a). These expenses are especially concerning given that farm wages have risen by over 8% in real terms since 2019 (USDA ERS, 2025).

Recent technological innovations offer solutions to these challenges that not only lower operational costs but also enhance sustainability by optimizing resource use across diverse farm settings. Although, the environmental implications of modernization extend beyond farm-level benefits. Improved water-use efficiency can decrease the impacts of drought, reduce groundwater depletion, and lower greenhouse gas emissions associated with energy-intensive irrigation (Robert W. Coates & Michael J. Delwiche, 2006). These gains are particularly critical in the Western U.S., where climate models project intensifying water scarcity and temperature extremes (Williams et al., 2022).

Despite irrigation advancements, the high upfront costs of adoption remain a critical obstacle, particularly for small- and medium-scale producers who lack access to capital (Deen et al., 2018). Financial constraints and uncertainty regarding economic returns have consistently been cited as key obstacles to the adoption of modern irrigation technologies (Koech et al., 2021). Furthermore, risk aversion due to unpredictable water allocations and production outcomes discourages farmers from investing in systems with long payback

periods (Fan & McCann, 2020; Rigby et al., 2010). Economic modeling indicates that investments in efficient irrigation often require multi-year commitments, with farm-specific payback periods influenced by crop value, local climate, and policy incentives (Cai & Rosegrant, 2004).

Despite extensive documentation of the technical, operational, and economic limitations of wheel-line irrigation systems in Western U.S. alfalfa production, there remains a significant gap in the literature regarding integrated decision-making frameworks that can guide producers through the process of irrigation modernization under conditions of drought and market uncertainty. Most existing studies either focus on the agronomic performance of irrigation technologies or provide broad economic analyses, but few offer a structured, practical approach that synthesizes these dimensions for small- and medium-scale producers. In particular, there is a lack of frameworks that help farmers and producers systematically evaluate the trade-offs between upfront investment, long-term operational savings, resource use efficiency, and risk factors such as water scarcity and price volatility. This chapter aims to address this gap by proposing a comprehensive economic framework, providing a foundation for future empirical research and practical decision support for producers considering irrigation system upgrades.

3.3 Economic Framework for Evaluating Wheel-Line Modernization

Given the complex interplay between technical limitations, economic constraints, and environmental pressures facing wheel-line irrigation systems in Western U.S. alfalfa production, a structured analytical framework is essential for evaluating modernization options. To address the gap in integrated decision-making tools for irrigation modernization,

this section develops an economic framework that combines cost-benefit analysis (CBA), risk assessment, and financial. The framework evaluates modernization strategies, such as low-pressure sprinklers, solar-powered automation, and soil moisture sensors, by quantifying their impacts on water efficiency, labor savings, energy use, and yield stability under drought and market volatility. By integrating these elements, the framework provides a decision-support tool for farmers considering irrigation system upgrades, policymakers evaluating conservation incentives, and researchers assessing sustainable agricultural practices.

3.3.1 *Cost-Benefit Analysis (CBA) Approach*

The economic assessment employs a comprehensive cost-benefit analysis framework specifically adapted for agricultural irrigation investments, building upon established methodologies in agricultural production economics (Kay & Edwards, 2017). This approach evaluates all quantifiable costs and benefits associated with modernizing wheel-line irrigation systems. Cost-Benefit Analysis evaluates the economic viability of a project using the formula:

$$Net\ Benefit = \sum_{t=0}^n \frac{B_t - C_t}{(1 + r)^t}$$

Where B_t represents benefits in period t (labor savings, energy reductions, yield improvements), C_t captures the total costs in year t (maintenance, operational adjustments), and r is the discount rate reflecting the opportunity cost of capital.

Scenario-based modeling is used to estimate payback periods, net present value, and sensitivity to changes in energy prices, labor rates, and water availability. This approach

enables producers and stakeholders to evaluate the financial implications of adopting new technologies under varying market and climate conditions.

3.3.2 Key Economic Metrics and Their Application

In addition to CBA, the economic evaluation of wheel-line irrigation modernization employs four principal financial metrics to determine investment viability. These metrics provide a framework for assessing both the absolute and relative economic merits of system upgrades while accounting for the time value of money and capital constraints characteristic of small and medium-sized farming operations. These metrics include:

- **Net Present Value (NPV):** NPV provides a comprehensive measure of an investment's value by discounting all future net benefits to their present value, thereby accounting for the time value of money and inflation. The general NPV formula applied in this context is:

$$NPV = \sum_{t=0}^T \frac{R_t}{(1+r)^t} - C_0$$

Where R_t is net return (benefit – costs) in year t , r is the discount rate reflecting the farm's cost of capital, and C_0 is initial investment cost.

A positive NPV indicates that the modernization project creates economic value exceeding its opportunity cost, while negative NPV suggests capital would be better deployed elsewhere. The discount rate applied in this analysis incorporates both the risk-free rate of return and appropriate risk premiums specific to agricultural investments in irrigation technology.

- **Internal Rate of Return (IRR):** Building on the NPV framework, the Internal Rate of Return (IRR) offers critical insight into investment efficiency by calculating the effective annual return rate:

$$IRR = \sum_{t=0}^n \frac{R_t}{(1+r)^t} - C_0 = 0$$

The IRR represents the discount rate that makes the NPV equal to zero, effectively showing the break-even return rate for the investment. This proves particularly valuable when comparing modernization options against alternative uses of capital or typical farm investment hurdles. A project with an IRR exceeding the farm's cost of capital (typically 8-12% for agricultural operations) would be considered economically attractive (Smith, 2013).

- **Benefit-Cost Ratio (BCR):** While NPV and IRR provide absolute measure of value creation, the Benefit-Cost Ratio (BCR) offers valuable complementary insights by establishing the relative efficiency of capital deployment. This proves particularly useful when comparing multiple modernization options or assessing projects under budget constraints:

$$BCR = \frac{\sum_{t=0}^T \frac{B_t}{(1+r)^t}}{\sum_{t=0}^T \frac{C_t}{(1+r)^t}}$$

where B_t represents the periodic benefits including labor savings, water saving, energy reductions, and yield improvements. C_t captures the total periodic costs, and r is the discount rate. A BCR exceeding unity indicates that each dollar invested generates more than one dollar in returns, with higher ratios signaling greater economic efficiency. This

metric proves particularly valuable when comparing alternative modernization scenarios or allocating limited capital across competing farm investments.

- **Payback Period:** Payback Period analysis addresses more immediate financial concerns that are often paramount for small and medium-sized operations. It measures the time required to recover the initial investment through annual net benefits. A shorter payback period generally indicates a more attractive investment, particularly in settings where liquidity and risk are significant concerns. The Payback Period is calculated as:

$$\text{Payback Period} = \min \left\{ T : \sum_{t=0}^T (B_t - C_t) \geq C_0 \right\}$$

Where B_t represents benefits in year t , C_t is Costs in year t , C_0 is initial investment, and T shows earliest year when cumulative net benefit is greater than initial investment. These four financial metrics provide a strong framework for evaluating the economic viability of modernizing wheel-line irrigation systems, capturing both the absolute and relative merits of investment under the capital constraints typical of small- and medium-sized alfalfa farms offer producers a clear basis for decision-making by quantifying the potential returns from labor, water, energy, and yield improvements. However, the economic feasibility of modernization is not without uncertainty, as factors such as fluctuating water availability, energy prices, and alfalfa market volatility, exacerbated by drought conditions as shown in Chapter 2, can significantly influence outcomes. To address these uncertainties and ensure the robustness of the investment decision, the following section introduces a risk assessment and sensitivity analysis, which evaluates how variations in key parameters affect the economic viability of modernization and informs strategies to enhance resilience under diverse climate and market scenarios.

3.3.3 *Risk Assessment and Sensitivity Analysis*

The risk assessment employs a scenario-based sensitivity analysis to evaluate the impact of variability in critical economic and environmental parameters on the financial metrics outlined in 3.3.2. This approach systematically tests the economic viability of modernization under diverse conditions, reflecting the uncertainties inherent in alfalfa production. The methodology involves the following steps:

- **Identify Key Risk Parameters:** Based on the cost-benefit framework, four primary variables are selected for sensitivity analysis including water prices, energy costs, alfalfa prices, and yield responses. These parameters are chosen due to their significant influence on modernization outcomes and their susceptibility to external shocks such as drought, market fluctuations, and policy changes.
- **Define Scenarios:** Three scenarios are constructed for each parameter: baseline (expected values), best-case scenario (favorable conditions), and worst-case scenario (adverse conditions) to capture a realistic range of outcomes. These scenarios are informed by historical data (e.g., USDA NASS, 2024; USDA ERS, 2025) and projections for water scarcity and energy price trends (EIA, 2025).
- **Simulate Economic Outcomes:** Using the cost-benefit analysis framework from the previous section, the simulation model recalculates financial metrics for each scenario, incorporating variations in the selected parameters.
- **Evaluate Robustness:** The robustness of modernization is assessed by examining the stability of financial metrics across scenarios, highlighting conditions under which investments remain viable or become unfeasible.

This approach ensures that producers can anticipate risks associated with climate variability and market fluctuations, aligning with the thesis's emphasis on resilience.

3.4 Economic Components of Irrigation Modernization: Costs and Benefits

A rigorous economic evaluation of irrigation modernization requires systematic identification and quantification of all relevant costs and benefits. This section details the principal economic components associated with transitioning from conventional to modernized wheel-line irrigation systems for alfalfa production in the Western United States.

3.4.1 Costs

The costs of modernizing wheel-line irrigation systems include both upfront and ongoing expenditures necessary to implement and maintain advanced technologies such as low-pressure sprinklers, solar-powered automation, and soil moisture sensors. These costs are categorized and described in Table 6.

Table 6.
Wheel line Irrigation Initial and Operating Costs

Cost Category	Description
Capital Investment	Upfront purchase and installation costs for new equipment (e.g., sprinklers, automation, sensors)
Maintenance	Annual expenses for repairs, servicing, and replacement parts
Labor	Changes in labor hours required for system operation and monitoring
Energy	Electricity or fuel costs for moving and operating wheel line systems
Water	Water purchase or pumping costs
Opportunity Cost	Foregone returns from alternative uses of capital or land

Water and energy costs are significant operational factors in evaluating the economics of wheel-line irrigation systems across Utah and other Western U.S. states. According to Utah State University Extension, water pumping costs are estimated at approximately \$70 per acre annually, depending on aquifer levels and system efficiency (John E. Keith, 2008; USU Extension, 2025). This estimate aligns with broader Western trends, where total water-related costs, including pumping, delivery, and water rights, typically range from \$66 to \$144 per acre annually, although smaller operations using on-farm surface or shallow groundwater sources may incur costs as low as \$25 per acre, depending on local conditions and infrastructure (Schaible & Aillery, 2012; USDA NRCS, 2023). Improved irrigation efficiency through system upgrades, such as low-pressure sprinklers and optimized scheduling, can reduce water application volumes and further lower costs. In addition, A 2022 USDA NRCS study from Montana shows that traditional wheel-line systems in the Western U.S. consume approximately 476 kWh per acre annually, compared to 286 kWh per acre for converted center-pivot systems (USDA NRCS, 2021)

3.4.2 Benefits

Modernizing wheel line irrigation systems can generate a range of direct and indirect benefits for alfalfa producers. These benefits include water savings, yield improvements, labor savings, energy savings, and risk reduction, each of which can be quantified to support the cost-benefit analysis (CBA) framework outlined in Section 3.3.1. These benefits are quantified below, supported by empirical data from field trials, peer-reviewed studies, and government reports.

- **Labor Savings:** Modernization, such as the adoption of automated movement and remote-control systems, can reduce this labor requirement to approximately 8 to 10 hours per acre per year (Fan & McCann, 2020). While exact savings depend on the level of automation and field configuration, multiple extension sources suggest that labor inputs can drop by 70–90%, especially when combined with automated timers and low-pressure sprinkler retrofits, which reduce maintenance needs and improve system efficiency (Cahn et al., 2013). At the current wage rate of \$18.43/hour, making these upgrades economically attractive in areas facing labor shortages and rising wage pressures. It can be estimated with the following formula:

$$\text{Annual Labor Saving (LS)} = (L_{\text{baseline}} - L_{\text{modern}}) \times W_{\text{labor}}$$

Where L_{baseline} is annual labor hours required under baseline system (conventional wheel-line), L_{modern} is annual labor hours required under modernized system, and W_{labor} is hourly wage rate (USD/hour).

- **Energy Savings:** Modernized systems often reduce energy use through more efficient pumps and lower-pressure operation. Field evaluations demonstrate that reducing pressure from 60 psi to 40 psi can decrease energy use without sacrificing irrigation performance. Based on a traditional energy use of 476 kWh per acre per year, lowering pressure by 20–30% can reduce energy consumption by 95 to 143 kWh per acre annually (Cahn et al., 2013; McGregor, 2023). Assuming an electricity rate of about 7¢ per kWh, this translates to annual energy savings of approximately \$7 to \$10 per acre. (EIA, 2025b; USDA NRCS, 2021). It can be estimated with the following formula:

$$\text{Annual Energy Saving (ES)} = (E_{\text{baseline}} - E_{\text{modern}}) \times P_{\text{energy}}$$

Where E_{baseline} is annual energy use under the baseline system (kWh or gallons), E_{modern} is annual energy use under modernized system (kWh or gallons), and P_{energy} is Price per unit of energy (USD/kWh or USD/gallon).

- **Water Savings:** Decreased water use resulting from improved distribution uniformity and reduced evaporation losses, contributing to both cost savings and enhanced drought resilience. Modernizing wheel-line irrigation systems by adopting low-pressure sprinklers and improved scheduling practices can lead to meaningful reductions in water use and cost. Traditional wheel-line systems typically apply between 3.5 and 4.5 acre-feet of water per acre per year, depending on crop, soil type, and local climate conditions (Stambouli et al., 2014). Field evaluations suggest that modernization can reduce water use by 0.5 to 0.8 acre-feet per acre, representing a 15–30% decrease in applied water through more uniform distribution and reduced evaporation losses (McGregor, 2023; USDA NRCS, 2023). At regional agricultural water prices ranging from \$20 to \$50 per acre-foot, these reductions translate into annual water cost savings of approximately \$70 to \$225 per acre, with most operations likely to save in the \$25 to \$50 per acre range (Hrozencik, 2025; USDA NASS, 2023). These savings improve operational efficiency while strengthening drought resilience in water-limited environments. It can be estimated with the formula:

$$\text{Annual Water Saving (WS)} = (W_{\text{baseline}} - W_{\text{modern}}) \times P_{\text{water}}$$

Where W_{baseline} is annual water use under the current system (acre-feet), W_{modern} is water use under the modernized system (acre-feet), and P_{water} is Price per acre-foot of water (USD).

- **Yield Improvements:** Modernized irrigation systems enhance crop yields by improving water application uniformity and timing, which reduces plant stress during critical growth stages. Field studies in the Pacific Northwest and California have shown that low-pressure sprinklers and automated irrigation can increase alfalfa yields by 0.2–0.4 tons per acre annually (a 5–9% increase) compared to conventional wheel-line systems (McGregor, 2023; Summers & Putnam, 2008). It can be estimated with the formula:

$$\text{Annual Yield Benefit (YB)} = (Y_{\text{modern}} - Y_{\text{baseline}}) \times A \times P_{\text{alfalfa}}$$

Where Y_{modern} is yield per acre with modernization (tons/acre), Y_{baseline} is yield per acre with baseline system (tons/acre), A is Irrigated acreage (acres), and P_{alfalfa} is Alfalfa price (USD/ton).

This increase not only boosts revenue but also stabilizes production under variable climatic conditions, addressing drought-induced yield declines noted in Chapter 2.

- **Risk Reduction:** Improved irrigation reliability may reduce the probability or severity of crop loss during drought or system failures. Adoption of more efficient irrigation technologies mitigates yield losses associated with drought and variable water supplies in Western U.S. agriculture (Schlenker et al., 2007). Similarly, some studies highlight that precision irrigation technologies enhance crop resilience by ensuring that water is delivered at optimal times, thereby reducing plant stress and stabilizing yields under adverse conditions (Hanson et al., 2007; Tubiello et al., 2007). The economic value of risk reduction can be expressed as the expected decrease in crop loss probability multiplied by the value of the crop at risk. Beyond direct economic benefits, reduced risk also provides psychological and operational advantages for producers, such as

greater confidence in production planning and reduced anxiety during drought events (Howden et al., 2007).

$$\text{Risk Reduction Benefit (RR)} = (\Delta P_{\text{loss}}) \times V_{\text{crop}}$$

Where ΔP_{loss} is reduction in probability of crop loss due to modernization, and V_{crop} is expected crop value at risk (USD/year). If the risk reduction is calculated over multiple years or scenarios, the expected value can be summed or averaged as appropriate.

- Total Benefit: The aggregate annual benefit from modernization is the sum of all individual benefit:

$$\text{Total Annual Benefit} = \text{WS} + \text{YB} + \text{LS} + \text{ES} + \text{RR}$$

By systematically measuring these costs and benefits, future researchers can construct a strong dataset for economic analysis. This structured approach ensures that all relevant and measurable benefits of irrigation modernization are incorporated into the economic evaluation framework, thereby supporting transparent and evidence-based decision-making for producers, researchers, and policymakers.

3.5 Data Collection and Simulation Model

3.5.1 Model Structure and Scenario Definition

The simulation framework is designed to integrate empirical data from wheel-line modernization studies, providing a flexible tool for evaluating both economic and agronomic outcomes across diverse regions and operational contexts. The model is structured to accommodate data from any relevant modernization project. As a case study, it is intended to incorporate results from Utah State University (USU) field studies

conducted between 2024 and 2026 in Cache, Davis, Iron, Duchesne, Wayne and Uintah counties. However, it is important to note that no primary field data from the USU project is available at this stage. Field research and data collection are ongoing, with results expected in future phases of the project. The framework is adaptable and can be updated as new data becomes available. Key variables included in the model are:

- Capital and Operating Costs: Equipment prices, installation, maintenance, lifespan, depreciation, sourced from manufacturer quotes or farm records.
- Labor Requirements: Hours per irrigation event or season for conventional vs. modern systems, measured via field observations.
- Energy and Water Inputs: Energy (kWh or fuel) and water use (acre-feet), via sensors, water balance methods, or utility records.
- Yield Impacts: Crop yield (e.g., 4.6 tons/acre for irrigated alfalfa), measured via field samples.

After collecting data, a simulation model compares baseline (conventional) and modernized scenarios, allowing users to input region-specific data (e.g., water prices, crop types) and forecast outcomes.

3.5.2 Guidance for Application and Implementation

The framework provides a step-by-step approach for researchers and stakeholders to evaluate wheel-line modernization:

- Define Scenarios: Specify baseline (conventional wheel-line) and modernized (e.g., low-pressure sprinklers, automation) scenarios, including technologies and crops.

- **Collect Data:** Gather data on costs (capital, maintenance, labor, energy, water), benefits (water savings, yield, labor, energy, risk reduction), and performance (e.g., uniformity) via field measurements, quotes, or records.
- **Apply CBA:** Calculate net benefits using enterprise budgets, tailoring to site-specific factors (e.g., water prices, soil types).
- **Compute Metrics:** Calculate NPV, IRR, BCR, and payback period, adjusting for taxes and subsidies.
- **Conduct Sensitivity Analysis:** Test variables (e.g., water prices, yields) under scenarios like drought or high energy costs.
- **Report Results:** Present findings in scenario comparison tables, documenting assumptions and limitations (e.g., data gaps, farm variability).

This rigorous, transparent framework equips stakeholders to assess the practical and financial case for wheel-line irrigation modernization in the Western U.S., capturing economic, operational, and risk-management dimensions essential for resilient agricultural decisions

3.6 Economic Value of Upgrading Wheel-Line Irrigation in Utah

To quantify the economic impact of modernizing wheel-line irrigation systems, this section estimates potential annual savings per acre and projects total statewide savings for Utah's alfalfa sector under three adoption scenarios. For each scenario, per-acre costs and benefits are obtained from Utah State University Extension budgets, the Congressional Research Service, and recent field studies, and then scaled to the statewide level. Using the cost-benefit analysis (CBA) methodology from Section 3.4, this section calculates costs, benefits, and financial metrics for each scenario, highlighting the framework's role in

supporting evidence-based decision-making in the context of Utah’s drought-prone agricultural landscape.

3.6.1 Traditional and Modernized Wheel-Line Systems

To quantify the economic benefits of modernizing wheel-line irrigation systems, Table 7 summarizes the estimated annual per-acre savings for Utah alfalfa producers in 2025. Modernizing wheel-line irrigation systems in Utah alfalfa production can conservatively generate annual per-acre economic benefits ranging from \$160 to \$278, with higher savings possible at operations with greater baseline costs or stronger yield response. Labor savings primarily driven by automation and reduced manual movement, typically account for the largest share, while energy and water savings further enhance operational efficiency. Yield improvements, valued at \$36 to \$71 per acre annually, provide an additional incentive and contribute to more stable revenue under variable climate conditions. These quantified benefits provide the foundation for the cost-benefit analysis presented in Section 3.3.1, supporting scenario modeling and statewide economic projections. The values are based on recent average alfalfa price (\$178/ton), farm wage rate (\$18.43/hour), electricity price (\$0.07/kWh), and for Utah producers (*EIA*, 2025a; *USDA NASS*, 2024).

Table 7.
Summary of Per-Acre Performance Metrics for Wheel-Line Modernization

Category	Wheel Line System		Notes
	Traditional	Modernized	
Labor	10–15 hours per acre (\$184–\$277)	2–4 hours per acre (\$37–\$74)	Based on wage rate

Category	Wheel Line System		Notes
	Traditional	Modernized	
Energy	476 kWh (\$33)	333–381 kWh (\$23–\$27)	20–30% reduction,
Water	3.5–4.5 ac-ft (\$70–\$225)	2.7–3.7 ac-ft (\$54–\$185)	0.5–0.8 ac-ft savings, \$20–\$50/ac-ft
Yield	3.4 tons per acre	+0.2–0.4 ton/acre (+\$36–\$71)	Slightly higher due to automation
Maintenance	\$15–\$30	\$20–\$40	Increase from improved uniformity
Total	\$298–\$494 (exclude capital)	\$105–\$191 (excl. capital, incl. yield)	Net annual savings: \$160–\$278/acre

Labor savings from modernizing wheel-line irrigation systems in Utah typically amount to a consistent 70–90% reduction in total labor hours, with annual requirements dropping from 10–15 hours per acre in traditional systems to just 2–4 hours after automation upgrades. Energy savings result from a 20–30% reduction in electricity use, modernized systems use 333–381 kWh per acre per year compared to 476 kWh for traditional systems, generating \$10–\$19 in annual cost reductions per acre based on prevailing electricity prices. Water savings range from 0.5–0.8 acre-feet per acre annually, with most operations achieving savings of \$25–\$50 per acre depending on local water rates and efficiency gains. Yield improvements, supported by Utah field studies, average 0.2–0.4 tons per acre annually and translate to an additional \$36–\$71 per acre at current alfalfa prices. While these categories form the basis for direct calculable net benefits (\$160–\$278 per acre per year), other indirect values, including risk reduction and lower probability of crop loss during drought, are not

explicitly included in the total but represent important supplementary advantages for producers adopting modernization.

3.6.2 Statewide Impact of Wheel-Line Modernization

Building on the per-acre economic benefits established in Section 3.4, this section projects the aggregate impact of wheel-line modernization for Utah's alfalfa sector under three adoption scenarios, including 10% (23,434 acres), 30% (70,302 acres), and 50% (117,170 acres) of alfalfa production on 234,339 acres in Utah using wheel line which is previously shown in Figure 14 (*USDA-NASS*, 2024), to assess the profitability and resilience benefits of modernizing wheel-line irrigation systems. Published data show that initial modernization costs for wheel-line irrigation systems range from \$295 to \$610 per acre (McGregor, 2023; USDA NRCS, 2023). Utah-specific field trials and extension surveys report an average cost of \$412 per acre for typical upgrades, including low-pressure sprinklers and basic automation (USU Extension, 2015). Actual costs can vary depending on the level of automation implemented and site-specific factors such as field size, slope, and terrain complexity. To provide a robust assessment of investment feasibility, this section recalculates statewide costs and benefits for a 15-year system lifespan using a literature-based capital investment range of \$295–\$820 per acre with inflation adjustment (*BLS*, 2025; USDA NRCS, 2023; Stubbs, 2016). Table 8 summarizes potential savings within the state, projecting annual benefits of \$3.75 million to \$32.57 million depending on adoption rates and savings scenarios.

Table 8.
Statewide Impact on Adoption Scenarios and Potential Savings

Adoption Rate	Acres Modernized	Annual Net Benefit (low)	Annual Net Benefit (Typical)	Annual Net Benefit (High)
10%	23,434	\$3.75 million	\$5.13 million	\$6.51 million
30%	70,302	\$11.25 million	\$15.40 million	\$19.54million
50%	117,170	\$18.75 million	\$25.66 million	\$32.57 million

Note. NPV and BCR use a 5% discount rate over 15 years Payback period is undiscounted. Annual maintenance costs for modernized systems (e.g., repair, sprinklers replacement) range from 1–3% of capital cost (USDA NRCS, 2023)

3.6.3 Scenario-Based Risk Assessment for Wheel-Line Modernization in Utah

A scenario-based risk assessment offers a clear analysis of how variations in key economic and environmental factors could influence the financial outcomes of wheel-line modernization in Utah. This assessment considers three scenarios, best case, base case, and worst case, that reflect a realistic range of outcomes for capital costs, annual net benefits, and long-term returns which are shown in Table 9.

The best-case scenario assumes optimal conditions such as favorable site factors and efficiencies, leading to rapid payback, very high benefit-cost ratios, substantial net present value, and exceptional returns that indicate a highly profitable and low-risk investment. The base case reflects average regional performance and costs, producing consistently strong returns including a reasonable payback period, solid benefit-cost ratio, positive net present value, and attractive internal rates of return. This scenario represents typical outcomes that many producers can expect under common field conditions. The worst-case scenario

assumes higher capital and maintenance costs alongside lower annual benefits, reflecting more challenging environments or complex operations. Despite these conservative assumptions, the investment remains financially viable with longer payback times and more modest but still positive returns that exceed the usual agricultural investment benchmarks. This underscores the resilience of the modernization approach, though careful cost and performance monitoring is suggested. Together, these scenarios illustrate that wheel-line modernization delivers financial benefits across a wide range of realistic economic and operational circumstances, supporting its attractiveness as an investment option. These projections underscore the significant economic and resource conservation benefits that can be achieved through targeted modernization of Utah’s wheel-line irrigation systems. Even at modest adoption rates, the resulting reductions in water use, labor demand, and energy consumption, together with modest yield improvements, offer a compelling case for both individual producers and policymakers. The initial investment required for widespread modernization is well within the scope of existing state and federal cost-share programs, further enhancing the feasibility and attractiveness of this strategy for advancing agricultural resilience and sustainability in Utah.

Table 9.
Different Scenario Outcomes of Modernized Wheel Line

Metric	Best Case	Base Case	Worst Case
Net Annual Benefit per acre	\$278	\$219	\$160
Capital Cost/acre	\$295	\$412	\$820
Annual Maintenance/acre	\$20	\$20	\$40
NPV (15 yrs, 5% discount)	\$2,879	\$2,116	\$973
Benefit-Cost Ratio (BCR)	10.76	5.14	1.87

Metric	Best Case	Base Case	Worst Case
Payback Period (years)	1.1	2.1	6.8
Internal Rate of Return	97%	37%	14%

3.7 Potential Challenges and Considerations

Despite the adaptability of this economic framework, several limitations and challenges must be considered. The accuracy of the analysis depends on the availability and quality of data on costs, yields, water and energy use, and labor requirements. Because the ongoing field studies are still in progress, early applications of the framework may be constrained by incomplete or inconsistent data. Additionally, economic outcomes are sensitive to fluctuations in water prices, energy costs, crop prices, and weather conditions. While scenario and sensitivity analyses are included to address such uncertainties, unexpected market or environmental shifts may still affect the robustness of the results. The framework is initially tailored to Western U.S. alfalfa systems and may require further adjustment to be fully generalizable to other regions, crops, or production systems. Furthermore, economic feasibility alone does not guarantee adoption, as producer decisions are also influenced by factors such as risk preferences, access to capital, and local policies. Finally, broader environmental and social impacts, such as groundwater recharge or community resilience, are not explicitly modeled, which may lead to an underestimation of the total benefits of modernization. Ongoing data collection, model refinement, and the integration of qualitative insights will be important to address these limitations as research progresses.

3.8 Discussion and Summary

This chapter has developed a comprehensive framework for evaluating the modernization of wheel line irrigation systems, with a primary focus on the economic feasibility and operational impacts relevant to small- and medium-scale farms in the Western United States, and introduces essential tools such as cost-benefit analysis, net present value (NPV), internal rate of return (IRR), and payback period to rigorously assess whether modernization investments yield net benefits over conventional practices.

The chapter identifies and details the core costs and benefits that must be measured to support robust economic analysis, including capital and operating expenditures, labor and energy savings, water use efficiency, and yield impacts. The analysis presented in this chapter demonstrates that modernizing wheel line irrigation systems offers substantial and quantifiable economic benefits for Utah's alfalfa producers, with implications for similar operations across the Western United States. Using a robust cost-benefit framework that incorporates capital investment (\$412 in average per acre), a 15-year system lifespan, and validated annual per-acre savings (\$160–\$278), we show that modernization consistently yields strong financial returns under a variety of adoption scenarios. Beyond direct economic gains, the analysis highlights additional operational and environmental advantages, including reduced labor dependency, improved water and energy use efficiency, and enhanced yield stability. These benefits are particularly salient in the context of rising input costs, ongoing labor shortages, and increasing climate variability. The chapter also acknowledges persistent challenges, such as the upfront financial barrier posed by modernization costs, variability in farm-level outcomes, and the need for accurate, site-specific data. However, the consistently favorable financial metrics across the literature-

based investment range suggest that modernization is a viable and resilient strategy for most small- and medium-scale producers. In summary, this chapter provides a rigorous, evidence-based foundation for both practical farm management decisions and policy development. The integrated analytical framework and standardized metrics developed here can guide future research, field trials, and incentive program design, ultimately supporting the broader adoption of efficient irrigation technologies and the long-term resilience of agriculture in drought-prone regions.

CHAPTER 4: CONCLUSION

4.1 Summary of Findings

This thesis has provided a comprehensive, multi-faceted assessment of the economic resilience of alfalfa production in the Western United States, focusing on the dual challenges of drought-induced price volatility and the role of modernized irrigation systems in enhancing sustainability. By integrating empirical market analysis with a rigorous economic feasibility framework, the research fulfills the objectives set forth in Chapter 1 which are quantifying the impact of drought and market factors on alfalfa price formation and evaluating the viability of upgrading wheel-line irrigation systems as a practical adaptation strategy.

Chapter 2 presented a detailed econometric analysis of alfalfa price dynamics across nine Western states from August 2021 to September 2024. The findings confirmed that drought severity, as measured by the Drought Severity and Coverage Index (DSCI), significantly increases alfalfa prices, especially when interacting with product quality and market variables. Forage quality, bale type, and feed market linkages (notably corn prices) emerged as primary drivers of price variation, while export demand and inventory levels further shaped local price outcomes. These results underscore the vulnerability of regional alfalfa markets to climate shocks and highlight the importance of adaptive management in sustaining economic viability.

Chapter 3 advanced the analysis by developing and applying a comprehensive economic framework to evaluate the modernization of wheel-line irrigation systems, which are widely used by small- and medium-scale alfalfa farms in the region. Incorporating cost-benefit analysis, financial metrics (e.g. NPV, IRR, BCR, and Payback Period), and risk assessment,

the framework was used to assess a range of modernization investments, including low-pressure sprinklers, solar-powered movers, and automation. The most recent analysis, based on a 15-year system lifespan and a literature-based capital investment range of \$295–\$820 per acre, demonstrates that modernization yields robust economic returns. The results highlight the economic viability of modernization and its potential to enhance resilience by reducing resource use and stabilizing yields under drought conditions.

Together, the findings from Chapters 2 and 3 illustrate that while drought continues to amplify price volatility and market stress for alfalfa producers, irrigation modernization offers a practical, evidence-based solution to mitigate these impacts. The research aligns with the thesis's original emphasis on understanding regional production dynamics and water scarcity challenges, confirming that alfalfa's economic importance in the Western U.S. requires adaptive, data-driven strategies for long-term sustainability. The insights from this study carry significant implications for policymakers, producers, and stakeholders aiming to improve agricultural resilience in the face of climate variability. The effect of drought on alfalfa prices suggests a need for enhanced risk management tools. Integrating drought indices like DSCI into crop insurance frameworks or disaster relief programs could better protect producers from income shocks, particularly during extreme events like the 2022 drought, which saw prices peak above \$350/ton. Moreover, the economic feasibility of irrigation modernization supports the case for targeted policy interventions. Subsidies, low-interest loans, or cost-sharing programs could offset the initial investment, accelerating adoption among capital-constrained small and mid-sized farms. Additionally, strategic forage reserves or export regulations could stabilize domestic supply during drought-induced shortages, addressing the limited buffering capacity of inventory levels.

For producers, the findings emphasize the value of proactive adaptation. Modernizing wheel-line systems can reduce operational costs (e.g., cutting labor and water use) while enhancing yield stability, a critical advantage given drought's price premium impact on premium grades. Farmers should also monitor feed market signals, such as corn price surges, to adjust production or marketing strategies, potentially diversifying feed sources to mitigate volatility. Extension services could further support adoption by providing real-time data and economic analyses tailored to individual farm conditions.

4.2 Suggestions for Future Work

While this thesis provides a foundation for understanding the economic dynamics of alfalfa markets and irrigation modernization, several limitations for future research remain. First, refining the price-formation model with finer spatial resolution (e.g., county-level data) could reveal localized drought effects obscured in state-level aggregates. Incorporating groundwater depletion metrics or soil moisture data alongside the DSCI might enhance precision, given the irrigation dependency noted in Chapter 2.

Also, longitudinal field studies are essential to validate modernization benefits over time. The Utah case study's 7% yield increase (average 0.3 tons/acre) and 15–30% resource savings need testing across multiple seasons and regions (e.g., Idaho, California) to confirm scalability. Comparative analyses of alternative technologies, like center pivots or drip irrigation could identify optimal solutions for diverse farm sizes and climates.

Furthermore, extending the analysis to other crops or developing dynamic tools, such as drought forecasts, price trends, and modernization outcomes could support real-time

decision-making. This would build on Chapter 2's market insights and Chapter 3's simulation approach, enhancing resilience across the Western agricultural landscape.

In conclusion, this research demonstrates that drought significantly disrupts alfalfa markets, yet modernized irrigation offers a viable path to resilience. By quantifying these dynamics and providing an actionable framework, the thesis contributes to sustainable agriculture in an era of climate uncertainty, fulfilling the objectives set in Chapter 1 and paving the way for impactful future inquiry.

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